

Function Survives Collapse

Admissibility, Entropy Flow, and the Geometry of Persistent Computation

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Abstract

Three results from distinct scientific domains converge on a single structural principle. In cryptography, the theory of additive randomized encodings demonstrates that arbitrary secure computation is achievable through anonymous additive aggregation: local provenance is irreversibly destroyed while computable functional invariants persist. In molecular biology, a study of transcription–translation coupling in *Escherichia coli* and related organisms demonstrates that gene expression is not a linear pipeline but a geometry-constrained coordination of partially decoupled flows, where mRNA fate depends critically on spatial accessibility rather than informational sequence alone. In plant developmental biology, a single-nucleus transcriptional atlas of early *Arabidopsis* seed development reveals that the embryo, endosperm, and seed coat coordinate their developmental programs not through full informational transparency but through low-dimensional signalling projections across symplastically isolated tissue interfaces. This essay argues that all three results are instances of a single deeper principle, formalized through the notion of admissible projection: a map $\pi : X \rightarrow M$ from a microscopic trajectory space X onto an admissible invariant manifold M , which preserves computable functionals while sacrificing microscopic recoverability under constrained entropy flow. The Relativistic Scalar-Vector Plenum framework supplies the overarching field-theoretic interpretation, with a variational Lagrangian formulation deriving the admissible projection as the attractor of constrained Euler–Lagrange flow. The framework is further extended through a categorical treatment of morphisms between admissible

systems, a renormalization-theoretic interpretation of scale-dependent projection, and a formal analysis of catastrophic projection collapse as the failure mode of robust systems. The central claim: successful computational, biological, and developmental systems do not preserve microscopic structure. They preserve low-dimensional functional invariants while continuously destroying the recoverability of the paths by which those invariants were reached.

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Introduction: Two Systems, One Collapse

Consider two facts from different sciences, presented without interpretation.

First: if n parties wish to compute a joint function of their private inputs without revealing anything beyond the output, they can each independently encode their input as a randomized group element and submit all encodings to an anonymous shuffle. The shuffler sums the encodings. The evaluator applies a decoder. The function output is recovered. No individual input is recoverable. Not approximately concealed — statistically indistinguishable from a transcript generated knowing only the output. This is the additive randomized encoding (ARE) framework. The foundational framework and its initial feasibility results appear in Halevi et al. (2023); the full universality result, establishing that statistical ARE exist for all functions, is due to Bitansky et al. (2025), who refuted the earlier conjecture that statistical security fails for simple functions. Efficient constructions remain available only for restricted circuit classes.

Second: in *Escherichia coli*, the enzyme RNase E is the primary driver of messenger RNA degradation. RNase E is not distributed uniformly through the cell. It is anchored to the inner membrane. Transcription occurs deep in the nucleoid. The geometric separation between these two regions means that nascent transcripts, still tethered to the RNA polymerase and threaded through the translation apparatus, are spatially inaccessible to the degradation machinery. Co-transcriptional degradation is therefore negligible — not because of an active protective mechanism, but because the geometry forbids the encounter. When RNase E is experimentally relocated to the cytoplasm, co-transcriptional degradation rises approximately sevenfold (Kim et al., 2026).

These phenomena appear to belong to incommensurable domains. One concerns privacy-preserving distributed computation. The other concerns the sub-cellular organization of bacterial gene regulation. Their vocabularies, methods, and epistemic standards have essentially no overlap.

Yet both are instances of the same structural situation: a system in which computable functional output persists under conditions that make microscopic reconstruction impossible. In the ARE case, impossibility is the design criterion. In the biological case, impossibility is a consequence of geometry. But in both cases the invariant that survives is low-dimensional relative to the space of microscopic configurations that could in principle produce it, and the mechanism of persistence is not protection of the path but indifference to it.

This convergence is not coincidental, and it is not merely analogical. Both

systems instantiate the same abstract structure: an irreversible projection from a high-dimensional microscopic trajectory space onto a low-dimensional admissible invariant manifold. The remainder of this essay develops that structure formally, applies it to each domain in detail, and argues that the Relativistic Scalar-Vector Plenum (RSVP) framework — a field-theoretic account of constrained dynamical systems developed for cosmological, cognitive, and computational applications — provides the natural overarching geometry for the unification.

The central thesis, stated once and carried through what follows: both systems preserve low-dimensional functional invariants by sacrificing microscopic recoverability under constrained entropy flow. The functional output is more stable than the path to it.

Structure of the Essay

Part I develops the core formalism: the admissible projection map, the notion of invariant persistence, and the role of entropy flow in destroying provenance while preserving output. Part II applies this to the ARE and shuffle-model MPC framework, showing how universal computation emerges from entropy mixing over locally randomized fragments. Part III applies it to transcription–translation coupling and mRNA degradation in bacteria, showing how spatial admissibility constraints govern which molecular trajectories can interact. Part IV develops the unified RSVP interpretation and introduces the field triple $(\Phi, \mathbf{v}, S)$ as its formal completion. Part V develops the generalization of privacy as accessibility geometry. Part VI examines constraint geometry, entropy as functional selector, and persistent computation without persistent representation. Part VII extends the framework toward a general theory of admissible systems and addresses the philosophical implications for microscopic ontology. The conclusion reflects on what it means for function to survive collapse.

Part I: Admissible Projection and Invariant Persistence

The Core Formalism

Let X denote a space of microscopic configurations or trajectories. In the cryptographic setting, X is the space of all possible joint inputs and randomness choices across n parties. In the biological setting, X is the space of all possible molecular trajectory ensembles: positions, velocities, binding states, and temporal sequences of every relevant molecular species. In both cases, X is vast, high-dimensional,

and inaccessible to direct observation.

Definition 1 (Admissible Invariant Manifold). *Let $\mathcal{F} = \{f_\alpha\}$ be a collection of computable functionals $f_\alpha : X \rightarrow Y_\alpha$ representing the observables of interest. The admissible invariant manifold M is the subspace of X , or of a suitable quotient thereof, on which the values of all f_α are well-defined and stable under small perturbations consistent with the constraints of the system. Formally,*

$$M = \{[x] \in X/\sim \mid f_\alpha(x) = f_\alpha(x') \text{ for all } x \sim x', \forall \alpha\},$$

where $x \sim x'$ if x and x' are indistinguishable under the constraint structure of the system.

Definition 2 (Admissible Projection). *An admissible projection is a map*

$$\pi : X \longrightarrow M$$

satisfying four conditions. First, functional preservation: $f_\alpha(\pi(x)) = f_\alpha(x)$ for all $x \in X$ and all α . Second, irreversibility: π is not injective; multiple distinct elements of X map to the same element of M . Third, entropy non-decrease: $H(\pi(X)) \leq H(X)$ in the sense that the projection increases or preserves the entropy of the induced distribution. Fourth, provenance destruction: there exists no efficient procedure for recovering x from $\pi(x)$ and the constraint specification of the system.

Remark. *The first condition ensures that the projection is not merely information loss: it is structure-preserving loss. The system discards everything about x except what is needed to compute the designated outputs. The second and fourth conditions together formalize what it means for microscopic recoverability to fail. The third condition locates this failure within a thermodynamic frame. The projection is not a filter that removes noise while preserving signal. It is a collapse that treats the microscopic trajectory as signal and the functional output as the invariant to be preserved.*

Entropy Flow and the Collapse of Provenance

The language of entropy flow is necessary here because neither paper under examination treats its subject as static. Both describe dynamic processes in which configurations evolve, interact, and are partially destroyed before a stable output is reached.

In the ARE setting, the relevant entropy flow is the injection of randomness by local encoders followed by aggregation. Each party P_i with input x_i samples

randomness r_i and produces an encoding $\hat{x}_i = \text{Enc}(x_i; r_i)$. The evaluator receives $\hat{y} = \sum_i \hat{x}_i$. The randomness injected by each encoder thermalizes the local representation: $\hat{x}_i$ carries no more information about x_i than is strictly necessary to reconstruct $f(x_1, \dots, x_n)$ from the aggregate.

In the biological setting, the relevant entropy flow is the diffusion, translation, and partial degradation of RNA transcripts across an inhomogeneous spatial field. The transcript does not persist as a stable object. It is a dynamical entity whose fate depends on which molecular encounters it undergoes, in which order, at which locations. The aggregate observable — protein production rate, mRNA lifetime, coupling ratio — is not a property of the transcript in isolation. It is a functional of the trajectory ensemble under the constraints imposed by the geometry of the cell.

In both cases, the functional output is more stable than the path to it. This is the operational content of admissible projection: M is an attractor, not an artifact. The trajectory is consumed. The invariant persists.

Pipeline Ontologies and Their Failure

Both domains under examination were historically understood through pipeline models. In theoretical computer science and distributed systems, the classical model of secure computation assumes that parties exchange messages through authenticated channels, that computations proceed in rounds, and that security is a property of the protocol architecture. The ARE framework challenges this picture fundamentally: it asks what can be computed when the only permitted global operation is addition — no rounds, no authenticated channels, no correlated setup. The answer, recently established, is: everything, in the statistical setting. The pipeline from sender to receiver is unnecessary. Coordinated entropy injection suffices.

In molecular biology, the classical Central Dogma — DNA $\rightarrow$ RNA $\rightarrow$ protein — carries implicit pipeline semantics: transcription produces a transcript, translation consumes it, the two processes are temporally and functionally sequential. The paper by Kim et al. (Kim et al., 2026) systematically dismantles this picture. Transcription and translation are not obligatorily coupled. Their degree of coupling depends on translation initiation rate. mRNA degradation does not simply follow transcript release; it depends on spatial accessibility to the degradation apparatus, which is itself a function of subcellular geometry and molecular identity. The pipeline from DNA to protein is not a pipeline. It is a partially coupled dynamical system whose observable outputs depend on constraint geometry rather

than sequential information flow.

The admissible projection framework offers a replacement for the pipeline ontology. Instead of asking how information travels from source to output, it asks which functionals of the source are preserved under the constraint-governed collapse of microscopic recoverability. That reorientation is the conceptual core of both results under examination.

Part II: Additive Randomized Encodings as Entropy Quotient Computation

The ARE Framework

The ARE model (Halevi et al., 2023) formalizes the following question: given an n -party function $f(x_1, \dots, x_n)$ and an Abelian group G , can each party P_i independently encode its input x_i into a group element $\hat{x}_i \in G$, such that the sum $\hat{y} = \sum_i \hat{x}_i$ allows recovery of $f(x_1, \dots, x_n)$ while revealing nothing else? A triple $\Pi = (\text{Setup}, \text{Enc}, \text{Dec})$ constitutes an ARE scheme if there exists a simulator Sim such that the distribution of $(pp, \hat{y})$ produced by the encoding procedure is indistinguishable from the distribution produced by $\text{Sim}(f(x_1, \dots, x_n))$ alone. The evaluator's view is simulatable from the output. No additional information is extractable.

Definition 3 (ARE Security). *An ARE scheme Π for f is statistically secure if there exists a simulator Sim such that for all $x_1, \dots, x_n$,*

$$\text{SD}(\text{Sim}(f(x_1, \dots, x_n)), \Pi(x_1, \dots, x_n)) \leq \delta(\lambda),$$

where δ is negligible and $\text{SD}(\cdot, \cdot)$ denotes statistical distance. The scheme is perfectly secure if $\delta = 0$.

In the language of Part I, the encoding map $x_i \mapsto \hat{x}_i$ is a local section of the admissible projection. The summation operator is the aggregation mechanism. The resulting $\hat{y}$ lives in M : it is the image of $\pi(x_1, \dots, x_n)$, containing exactly the information needed to decode f and nothing more.

The OR Construction as Phase Transition

The simplest nontrivial ARE is the scheme for the OR function over n binary inputs (Halevi et al., 2023). Each party encodes $x_i = 0$ as the identity element $0 \in G$

and $x_i = 1$ as a uniformly random element $g_i \in_R G$. The sum $\hat{y} = \sum_{i=1}^n \hat{x}_i$ is zero when $\text{OR}(x) = 0$ and uniformly random in G (with probability $1 - |G|^{-1}$) when $\text{OR}(x) = 1$.

The evaluator decodes: output 0 if $\hat{y} = 0$, output 1 otherwise. Security is perfect when $\text{OR}(x) = 1$: the transcript $\hat{y}$ is indistinguishable from random and carries no information about which party contributed the active input, how many parties were active, or what the individual inputs were.

This construction instantiates a phase transition in the aggregate state. The presence of a single active participant thermalizes the sum. One unit of entropy injection is sufficient to mask the entire microscopic configuration. The aggregate observable shifts from a stable fixed point — the identity element — to a uniformly distributed random variable. The trajectory space X for OR contains 2^n configurations. The admissible manifold M has two elements: the identity and the uniform distribution over G . The projection $\pi : X \rightarrow M$ maps all configurations with any active party to the same distributional output. Provenance is destroyed by the projection. The functional output is preserved exactly.

Universality Through Entropy Amplification

The deeper result, established by Bitansky et al. (2025) and refuting the earlier conjecture that statistical ARE might fail even for simple functions like equality over small domains, is that statistically secure AREs exist for all functions, without efficiency guarantees for general functions.

The construction proceeds in two conceptually distinct stages. First, a *leaky* ARE (LARE) is constructed for any finite function. A LARE satisfies a relaxed security condition governed by a specific leakage function:

$$\ell(x, y) = \begin{cases} x & \text{if } f(x, y) = 1, \\ \perp & \text{if } f(x, y) = 0. \end{cases}$$

That is, when the output is 1, the encoding may reveal the first party's input x while concealing the second party's input y entirely; when the output is 0, nothing leaks beyond the output. The LARE construction encodes party 1's input x as a randomized vector with a zero in coordinate x and random group elements elsewhere, and party 2's input y as a randomized vector with zeros precisely at coordinates where $f(\cdot, y) = 1$. The aggregate sum therefore has a zero in position x if and only if $f(x, y) = 1$, yielding correct decoding and the specified leakage profile. This is perfectly leaky-secure and correct up to a $O(|G|^{-1})$ error.

Second, the LARE is lifted to a full ARE through secret sharing and amplification. For a security parameter τ , party 1 samples τ additive secret shares $\mathbf{x}_1, \dots, \mathbf{x}_\tau$ of the unit vector e_x , and party 2 samples τ shares $r_1, \dots, r_\tau$ of 0. The parties run τ parallel LARE instances for the inner-product-with-mask function $g(\mathbf{a}, (\mathbf{b}, r)) = \langle \mathbf{a}, \mathbf{b} \rangle + r$, obtaining outputs $\langle \mathbf{x}_i, f_y \rangle + r_i$ for each i . Their sum recovers $\langle e_x, f_y \rangle + 0 = f(x, y)$. Security follows because with probability $1 - 2^{-\tau+1}$ at least one share r_i ensures the corresponding output is 0, at which coordinate nothing leaks; whenever the output is 1, the leaked share $\mathbf{x}_i$ is uniformly random and independent of the actual input x .

For any two-party function $f : D_1 \times D_2 \rightarrow \{0, 1\}$, the reduction to inner product is achieved by representing party 1's input x as the unit vector e_x and party 2's input y as the truth-table vector f_y encoding all values $f(\cdot, y)$:

$$f(x, y) = \langle e_x, f_y \rangle.$$

Function evaluation becomes geometric projection. Computation reduces to inner product, and inner product reduces to the LARE gadget. Extension to multi-party functions proceeds via decomposable randomized encodings and oblivious transfer, yielding efficient constructions for NL/poly and, assuming one-way functions, for all of P/poly .

Proposition 1 (Bitansky et al. 2025, Theorem 1.1). *For any k -party function $f : D^k \rightarrow D$ over a finite domain and any error bound $\varepsilon > 0$, there exists an ARE for f with statistical correctness and security errors at most ε . Moreover, any function $f \in NL/\text{poly}$ admits a polynomial-time efficient ARE with statistical correctness and security errors $2^{-\tau}$, and any function $f \in P/\text{poly}$ admits a computationally secure ARE of size $O(\tau \cdot S)$ where S is the circuit size of f , using only a one-way function.*

The efficiency cost is real and large. But the existence result is philosophically decisive: the only global operation required is addition. All coordination is accomplished through local entropy injection and the accumulation of admissible invariant structure in the aggregate. The pipeline from sender to receiver does not merely disappear; it is shown to have been unnecessary.

The Shuffle as Entropy Quotient Map

In the shuffle model of distributed computation, the architectural primitive is not a channel between parties but a mechanism that receives all messages and delivers them in a uniformly random permutation, erasing sender identity. The ARE

framework interfaces with the shuffle model as follows: the ARE messages $\hat{x}_i$ are submitted to the shuffle, which sums them while destroying all record of which message came from which sender (Halevi et al., 2023).

This shuffler is naturally interpreted as an entropy quotient map. Let Σ be the set of all sender-labeled message multisets. The shuffle defines an equivalence relation on Σ : two labeled multisets are equivalent if they produce the same unlabeled aggregate. The quotient $\Sigma/\sim$ is the image of the shuffle. The map $\sigma : \Sigma \rightarrow \Sigma/\sim$ destroys provenance — the sender labels — while preserving the additive invariant, the sum.

In the language of Part I, σ is an instance of admissible projection. The fiber $\pi^{-1}(\hat{y})$ over any aggregate value $\hat{y}$ contains all possible sender assignments consistent with that aggregate. These fibers are arbitrarily large; the evaluator cannot determine which fiber element was the actual configuration. The functional output, recovered by applying Dec to $\hat{y}$, is fiber-invariant: it does not depend on which element of $\pi^{-1}(\hat{y})$ was realized.

Remark. *The cryptographic security condition is equivalent to the geometric statement that the functional observable is constant on fibers of π . This is the core content of simulation-based security, reexpressed in the language of admissible projection. No simulator is required to reconstruct the fiber element, the actual inputs. It need only reproduce the fiber-invariant observable, the output distribution.*

Robust ARE and the Geometry of Residual Leakage

The 2023 paper introduces a strengthened notion: *robust ARE*, in which security holds not only against an external evaluator but against coalitions of parties who may pool their encodings to learn the partial sum contributed by the remaining honest parties. In robust ARE, the security guarantee is that the partial sum $\hat{y}_H = \sum_{i \in H} \hat{x}_i$ reveals no more than the *residual function* $f_{H,x}$: the function f with the inputs of all honest parties fixed. Colluding parties can compute all completions of the function, but they cannot learn the individual inputs of the honest parties beyond what the residual function forces.

This notion has a natural interpretation in obstruction geometry. The information the adversary cannot extract corresponds to a class recording the failure of local-to-global extension. The colluding parties have local information — their own inputs and the partial sum — but cannot extend it to a global reconstruction of the honest parties' inputs. The obstruction to this extension is precisely the entropy injected by the honest parties' local encoders. The topology of the hon-

est parties' encoding space prevents the adversary from reconstructing a global section from local data.

Part III: Transcription, Coupling, and Spatial Admissibility

The Geometry of Bacterial Gene Expression

The paper by Kim et al. (Kim et al., 2026) investigates the degree to which transcription, translation, and mRNA degradation are co-temporally coupled in bacteria. The received view holds that these three processes are tightly integrated: the ribosome follows closely behind the RNA polymerase, translating the transcript as it emerges, while the degradation machinery stands ready to process released transcripts promptly.

The paper systematically undermines this picture. Its central finding is that transcription–translation coupling is not a universal feature of bacterial gene expression but a condition-dependent outcome governed by translation initiation rate. Genes with weak ribosome binding sites exhibit significant premature transcription termination — the RNA polymerase terminates early because no ribosome is close enough to prevent it — while genes with strong ribosome binding sites maintain tight coupling. The progression factor, defined as the ratio of downstream to upstream transcript coverage, correlates strongly with translation efficiency genome-wide, with the correlation coefficient $r = -0.93$ for the engineered variant panel and a genome-wide separation of high-PF and low-PF gene classes upon Rho inhibition.

This is a quantitative demonstration that the pipeline model of bacterial gene expression is wrong. There is no universal coupling. There is a coupling field whose local value depends on the kinetic accessibility of the ribosome to the nascent transcript.

Membrane Localization as Topological Obstruction

The most striking single result in the paper, from the perspective of admissible projection, is the role of RNase E localization. RNase E, the primary endoribonuclease responsible for mRNA degradation in *E. coli*, is anchored to the inner membrane via its membrane-targeting sequence. Transcription occurs in the nucleoid, which is spatially separated from the inner membrane. This geometric separation creates a *spatial admissibility constraint*: the trajectories of nascent transcripts and the trajectories of membrane-bound RNase E are largely non-intersecting.

Let Γ_{RNase} denote the spatial trajectory ensemble of RNase E molecules and Γ_{nascent} denote the spatial trajectory ensemble of nascent transcripts still tethered to the RNA polymerase in the nucleoid. The paper’s central finding can be stated geometrically as

$$\Gamma_{\text{RNase}} \cap \Gamma_{\text{nascent}} \approx \emptyset.$$

This intersection emptiness is not an active protective mechanism. No molecular machinery enforces the separation. The geometry of the cell — the localization of the membrane, the compaction of the nucleoid, the diffusion rates of the relevant molecules — produces the separation as a consequence of spatial organization.

The experimental confirmation is decisive. When RNase E is relocated to the cytoplasm by deleting its membrane-targeting sequence, the co-transcriptional degradation rate k_{d1} rises from $0.042 \pm 0.060 \text{ min}^{-1}$ to $0.31 \pm 0.08 \text{ min}^{-1}$, an approximately sevenfold increase (Kim et al., 2026). The functional invariant — negligible co-transcriptional degradation — was maintained not by any biochemical protection mechanism but by the topological separation of admissible trajectory spaces.

Definition 4 (Spatial Admissibility Constraint). *A spatial admissibility constraint on a molecular interaction is a condition of the form*

$$\Gamma_A \cap \Gamma_B = \emptyset \pmod{\epsilon},$$

for molecular species A and B , where Γ_A and Γ_B are their respective spatial trajectory ensembles and ϵ is a threshold reflecting diffusion timescales relative to the dynamical timescale of the interaction being considered. The constraint renders the interaction kinetically inaccessible regardless of thermodynamic favorability.

Translation Efficiency as Coupling Field

The genome-wide analysis in Kim et al. (2026) establishes that translation efficiency — a proxy for translation initiation rate per transcript — is the primary determinant of transcription–translation coupling across the *E. coli* genome. Among the 1,336 genes analyzed, high-TE genes show progression factors near 1, reflecting full coupling, while progressively lower TE values associate with progressively lower progression factors, corresponding to increased premature termination.

This finding transforms TE from a purely informational quantity — a rate of protein production — into a field variable governing the dynamical coherence

of the transcription–translation system. High TE maintains the RNAP–ribosome distance below the threshold of approximately 80 nucleotides at which Rho factor loading and premature termination become probable. Low TE allows the RNAP to outpace the ribosome, opening the spacer, enabling Rho loading, and driving the system into a premature termination regime.

In the admissible projection language, TE parametrizes the projection π . High-TE genes are projected onto an admissible manifold M_{coupled} in which full-length transcript production and translation are coordinated. Low-TE genes are projected onto $M_{\text{uncoupled}}$, a different region of the invariant manifold characterized by partial transcript release, altered degradation kinetics, and two-phase mRNA decay signatures. The transition between these regimes is continuous rather than sharp: TE acts as a tunable coupling constant for the field governing spatial and temporal coordination between the molecular flows.

Premature Termination as Trajectory Collapse

The phenomenon of premature transcription termination is especially illuminating through the lens of admissible projection. In a strongly coupled gene, the RNAP–ribosome complex produces a coherent joint trajectory. The transcript is never freely released into the cytoplasm during transcription; it remains tethered to both the polymerase and the ribosome. This coupled trajectory is protected from premature termination and, as argued above, from co-transcriptional degradation by spatial separation from RNase E.

In a weakly coupled gene, the RNAP outpaces the ribosome. The spacer RNA becomes accessible to the Rho helicase. Rho loads onto the exposed transcript and drives termination. The nascent transcript is released prematurely — incomplete, lacking the 3' UTR or intrinsic terminator stem-loop structure, and now diffusing freely in the cytoplasm.

This released transcript occupies a different region of X . Its spatial trajectory ensemble has changed from Γ_{nascent} , tethered and nucleoid-proximal, to Γ_{free} , diffusing and cytoplasm-accessible. Critically, Γ_{free} is no longer disjoint from $\Gamma_{\text{RNase E}}$. The premature transcript becomes accessible to membrane-localized RNase E through normal diffusion and is rapidly degraded. The apparent increase in the apparent co-transcriptional degradation rate k_{d1} observed in weak-RBS strains is therefore not genuinely co-transcriptional. The paper carefully distinguishes k_{d1}^* — true co-transcriptional degradation of RNAP-tethered transcripts — from k_{d1} , the apparent degradation rate during the transcription window, which includes decay of prematurely released transcripts. The correlation $r = 0.94$ between pre-

mature termination probability and k_{d1} confirms that the elevated apparent rate is a consequence of trajectory collapse: the transcript exits the admissible coupled trajectory and enters a new trajectory ensemble with different accessibility and degradation kinetics.

Species-Specific Admissibility Topologies

One of the philosophically most significant aspects of Kim et al. (2026) is its comparative analysis across three bacterial species with fundamentally different sub-cellular organizations.

In *E. coli* and *Bacillus subtilis*, the primary ribonuclease — RNase E and its functional homologue RNase Y, respectively — is membrane-localized. In both species, co-transcriptional degradation is negligible for standard cytoplasmic protein-encoding genes: $k_{d1} \approx 0.042 \text{ min}^{-1}$ for *E. coli* lacZ and $k_{d1} \approx 0.025 \text{ min}^{-1}$ for *B. subtilis* lacZ, both far below the corresponding post-transcriptional rates. The spatial admissibility constraint $\Gamma_{\text{RNase}} \cap \Gamma_{\text{nascent}} \approx \emptyset$ holds in both organisms as a consequence of membrane localization.

In *Caulobacter crescentus*, the situation is fundamentally different. RNase E in this organism and its relatives is cytoplasmic. The paper observes rapid co-transcriptional degradation in *C. crescentus*: $k_{d1} \approx 1.3 \text{ min}^{-1}$, approximately an order of magnitude higher than in *E. coli*. The spatial admissibility constraint has collapsed. Γ_{RNase} now overlaps substantially with Γ_{nascent} , making co-transcriptional degradation accessible and prevalent.

Remark. *This comparison establishes that the admissibility topology is not a universal feature of bacterial gene expression but a species-specific geometric property determined by the subcellular localization of the primary degradation machinery. The same functional invariant — controlled gene expression — survives in all three organisms. The microscopic mechanism by which it is achieved differs entirely. Identical informational goals are realized through distinct geometric accessibility structures. This is better understood not merely as different evolutionary solutions to the same problem, but as distinct admissibility geometries realizing the same invariant class.*

The three species thus instantiate three distinct admissible projection geometries for the same functional task. This is not evolution finding three paths to the same solution. It is three different implementations of the same abstract admissible projection $\pi : X \rightarrow M$, with different constraint structures on X producing different intermediate trajectory ensembles while converging on the same class

of invariant observables.

Part IV: The RSVP Framework and the Unified Geometry

The Field Triple

The Relativistic Scalar-Vector Plenum (RSVP) framework posits that the relevant structure of any sufficiently complex dynamical system is captured by a field triple $(\Phi, \mathbf{v}, S)$, where $\Phi : \mathcal{D} \rightarrow \mathbb{R}$ is a scalar field representing the density or intensity of admissible configurations at each point of the domain $\mathcal{D}$; $\mathbf{v} : \mathcal{D} \rightarrow T\mathcal{D}$ is a vector field representing the flow of trajectories across the domain; and $S : \mathcal{D} \rightarrow \mathbb{R}$ is an entropy field representing the local degree of microscopic indistinguishability or randomization.

The dynamics of RSVP systems are governed by the interplay of these three fields. Regions of high Φ are regions of high admissibility: trajectories that reach these regions are likely to produce stable functional outputs. Regions of high S are regions of high entropy: microscopic recoverability is low, and the projection π is far from injective. The vector field $\mathbf{v}$ governs how trajectories flow from high-recoverability regions toward high-entropy, high-admissibility regions of M .

A schematic constraint governing the field triple can be written as

$$\partial_t \Phi + \nabla \cdot (\Phi \mathbf{v}) = -\lambda S,$$

expressing the condition that admissibility density is transported by the trajectory flow $\mathbf{v}$ and dissipated at a rate proportional to local entropy production. Regions where entropy increases rapidly are regions where admissibility is consumed. Stable function requires that Φ remains above a threshold $\Phi_{\min}$ even as S increases.

This equation can be derived as the Euler–Lagrange equation of an action functional for the field triple. Define the RSVP Lagrangian density

$$\mathcal{L}(\Phi, \mathbf{v}, S) = \frac{1}{2} |\mathbf{v}|^2 - V(\Phi) - \alpha S + \beta \Phi \nabla \cdot \mathbf{v},$$

where $V(\Phi)$ is a potential penalizing configurations below $\Phi_{\min}$, $\alpha > 0$ weights the entropic cost of local randomization, and $\beta > 0$ couples the admissibility field to the divergence of the trajectory flow. Variation with respect to $\mathbf{v}$ yields the flow equation relating trajectory velocity to the gradient of the admissibility potential; variation with respect to Φ yields the conservation law stated above; and variation with respect to S yields the condition that entropy production is

bounded by the coupling coefficient α . The admissible manifold M emerges as the critical-point locus of this system under the constraint that S is non-decreasing along trajectories of $\mathbf{v}$. This variational formulation makes the RSVP framework a genuine geometric field theory rather than a merely interpretive framework: the admissible projection $\pi : X \rightarrow M$ is not postulated but derived as the attractor of the constrained Euler–Lagrange flow.

Definition 5 (RSVP Admissible Projection). *In the RSVP framework, the admissible projection $\pi : X \rightarrow M$ is the asymptotic map*

$$\pi(x) = \lim_{t \rightarrow \infty} \phi_t(x),$$

where ϕ_t is the flow of $\mathbf{v}$ on $\mathcal{D}$, restricted to $\{\Phi \geq \Phi_{\min}\}$ and conditioned on entropy non-decrease along trajectories. The manifold M is the attractor of this restricted flow.

This recovers the admissible projection of Part I as the long-time limit of constrained flow. The trajectory x is not preserved; what is preserved is the functional output of the limiting point $\pi(x)$ on the admissible manifold.

ARE as RSVP Field Dynamics

In the ARE setting, the RSVP triple interprets as follows. The scalar field Φ encodes the probability density of ARE encodings over the group G . High- Φ regions correspond to encodings consistent with valid ARE transcripts: their aggregate lies in the range of the decoder and produces a valid function output. The vector field $\mathbf{v}$ is the flow induced by the aggregation operation. Individual encodings $\hat{x}_i$ flow toward their aggregate $\hat{y} = \sum_i \hat{x}_i$. This flow is constrained to preserve the functional invariant $f(x_1, \dots, x_n)$ as a stable attractor. The entropy field S captures the randomness injected by local encoders. High- S regions of the encoding space are regions where many different joint input configurations map to the same aggregate: these are the large-fiber regions of π , corresponding to strong provenance destruction.

The security condition of ARE — that the evaluator’s view is simulatable from the output alone — is the RSVP condition that S is high in the image of π . Many microscopic configurations are consistent with any observed aggregate, and no efficient procedure can distinguish them. Security and entropy are here not in tension. They are the same phenomenon viewed from different angles.

Bacterial Gene Expression as RSVP Field Dynamics

In the bacterial gene expression setting, the RSVP triple interprets as follows. The scalar field Φ encodes the spatial density of admissible configurations. High- Φ regions are regions of the cell where the relevant molecular interactions — productive transcription–translation coupling, RNase-transcript encounter — can occur. The nucleoid interior and the membrane periphery define distinct high- Φ regions for different processes. The vector field $\mathbf{v}$ encodes the flow of molecular trajectories: RNAP movement along DNA, ribosome translocation along mRNA, transcript diffusion through the cytoplasm, RNase E diffusion along the membrane. The entropy field S captures spatial and configurational randomness. High- S regions are regions of the cytoplasm where many different molecular trajectory histories produce the same observable output. Low- S regions are regions of tight coupling where the trajectory is constrained to follow a specific path, as in the RNAP–ribosome coupled complex in a high-TE gene.

The transition from coupled to uncoupled transcription, governed by translation efficiency, is a transition in the RSVP sense: a change in the local structure of $\mathbf{v}$ as the coupling flow becomes non-coherent, an increase in S as the spacer RNA becomes accessible to multiple degradation pathways, and a shift of the trajectory from a low-entropy path on M_{coupled} to a higher-entropy path on $M_{\text{uncoupled}}$. The admissibility field Φ remains high in both regimes, since functional regulation is achieved in either case, but the trajectory by which the invariant is approached changes fundamentally.

Obstruction, Cohomology, and the Topology of Non-Interaction

The two central examples of admissible projection in this essay are geometrically dual to each other in an important sense. In the ARE case, the projection destroys provenance by mixing. The entropy injected by local encoders thermalizes the aggregate, making the fiber of π large. The functional invariant survives not because anything protects it from noise, but because it is a stable functional of the aggregate that remains well-defined regardless of which fiber element was realized.

In the bacterial case, the projection preserves function by separation. The spatial admissibility constraint prevents certain molecular encounters from occurring, thereby preventing certain degradation trajectories from being realized. The functional invariant survives not because anything actively protects transcripts from RNase, but because the geometry of the cell ensures that the relevant trajec-

tories do not intersect.

These two mechanisms — mixing and separation — are dual in the sense of obstruction cohomology. Mixing corresponds to a cohomological trivialization: the fiber of π is so large that no distinguisher can recover a global section from local data. The extension problem — recover x from $\pi(x)$ — has no solution because too many solutions are consistent with the observed data. Separation corresponds to a cohomological obstruction: the extension problem — allow RNase to reach the nascent transcript — has no solution because the geometry prevents the required paths from existing. The relevant cochains cannot be assembled into a global section.

Both mechanisms prevent the recovery of microscopic trajectory information. Both allow functional invariants to be computed and preserved. They are two realizations of the same abstract phenomenon: the impossibility of local-to-global extension in a constrained topological setting.

Life, Computation, and the RSVP Attractor

The deepest unification offered by the RSVP framework is the identification of M as an attractor of the dynamical system defined by the field triple. In both the cryptographic and biological settings, the relevant systems are not designed to reach M in finite time and then halt. They are ongoing dynamical processes in which the admissible manifold is continuously approached, partially realized, and partially escaped. ARE encodings are not computed once; they are produced by parties who participate in many future computations. Transcripts are not produced in isolation; each transcript is one element of a continuous flux of production and degradation that maintains cellular protein levels in a dynamical steady state.

In both cases, the attractor M is not a fixed point but a statistical regime: a region of the state space that the system spends most of its time near, and to which it returns after perturbation. The functional invariants preserved by π are not static properties of a configuration; they are statistical properties of the ensemble of configurations realized over time. A system computes, or a cell regulates, not by instantiating a fixed microscopic configuration, but by maintaining a dynamical ensemble of microscopic trajectories whose time-averaged projection $\pi(x(t))$ remains concentrated on the admissible manifold M . Function is not a property of any single trajectory. It is a property of the projection of a trajectory ensemble onto an admissible invariant manifold under constrained entropy flow.

Part V: Provenance, Privacy, and the Topology of Accessibility

Privacy as Accessibility Geometry

Classical notions of privacy are framed negatively. A system is private if an observer cannot infer a hidden variable, cannot distinguish between two neighboring databases, or cannot reconstruct a secret from observable outputs. These formulations treat privacy as an epistemic limitation imposed upon an observer. The admissible projection framework developed in this essay suggests a different interpretation. Privacy is not fundamentally about secrecy. It is about accessibility geometry.

In the ARE setting, the evaluator fails to reconstruct the microscopic inputs not because the inputs are hidden behind an encrypted wall, but because the admissible projection $\pi : X \rightarrow M$ collapses many microscopic configurations into the same observable invariant. The evaluator is trapped on the quotient manifold. The inaccessible information is inaccessible because the topology of the fibers forbids unique reconstruction.

In the bacterial setting, the same structure appears physically. RNase E cannot access the nascent transcript because the trajectories required for encounter do not exist within the admissible geometry of the cell. The transcript is not encrypted. It is spatially inaccessible. No key is needed to protect it because no path leads to it.

Definition 6 (Topological Privacy). *A system exhibits topological privacy with respect to an observable $f : X \rightarrow Y$ if the admissible projection $\pi : X \rightarrow M$ induces fibers $\pi^{-1}(m)$ whose internal structure is inaccessible to an observer constrained to trajectories within M . The inaccessibility need not be cryptographic; it may arise because the admissible geometry forbids the trajectories required for reconstruction.*

Under this interpretation, privacy becomes a property of the topology of accessibility rather than merely a property of information encoding. The abstract structure appears identically in shuffle-based computation, in membrane-localized degradation, in thermodynamic coarse-graining, and in cognitive abstraction. In each case the observer or agent is restricted to a low-dimensional projection manifold whose geometry forbids reconstruction of the microscopic trajectory ensemble.

Fiber Structure and Functional Equivalence Classes

The fibers of π play a central structural role throughout this essay. They are the equivalence classes of microscopic trajectories that produce the same admissible invariant. In the ARE setting, the fiber over a transcript $\hat{y}$ is

$$\pi^{-1}(\hat{y}) = \{(x_1, \dots, x_n, r_1, \dots, r_n) \mid \sum_i \text{Enc}(x_i; r_i) = \hat{y}\}.$$

This fiber is enormous. Many distinct joint inputs and randomness assignments produce the same aggregate transcript. Security follows from the inability to distinguish which fiber element was realized.

In the biological setting, a measured protein abundance level may correspond to many distinct transcription initiation histories, many distinct ribosome loading sequences, many distinct degradation trajectories, and many distinct diffusion paths. The observable survives because it is fiber-invariant: its value is the same regardless of which microscopic realization within the fiber was actually traversed.

Proposition 2 (Invariant Stability Principle). *The stability of a functional observable increases as its dependence on fiber-internal structure decreases. An observable constant across large fibers of π remains stable under substantial microscopic variation.*

This principle is immediate. Any observable depending sensitively on microscopic trajectory detail will fluctuate strongly under perturbation of the fiber structure. The principle explains simultaneously why shuffle MPC remains secure under random permutation, why biological regulation tolerates molecular stochasticity, why thermodynamic observables remain stable under microscopic chaos, and why semantic cognition tolerates representational variation. The observable survives because it is a property of the equivalence class, not of any individual trajectory.

The Algebra of Admissible Observables

The fiber structure of π determines not merely which observables are stable but which observables are *definable* on the quotient manifold M . This observation motivates a precise characterization of the admissible observable algebra.

Definition 7 (Admissible Observable Algebra). *Let $\pi : X \rightarrow M$ be an admissible*

projection. The algebra of admissible observables is the set

$$\mathcal{O}(M) = \{f : X \rightarrow \mathbb{R} \mid f(x) = f(x') \text{ whenever } \pi(x) = \pi(x')\}.$$

Elements of $\mathcal{O}(M)$ are precisely the functions on X that factor through π : they are constant on fibers and therefore well-defined on the quotient.

This definition has immediate structural content. The algebra $\mathcal{O}(M)$ is closed under pointwise addition, multiplication, and composition with continuous functions, making it a genuine commutative algebra. Any measurement performed on a system governed by the projection π that produces a stable, reproducible result must belong to $\mathcal{O}(M)$; measurements depending on fiber-internal structure cannot be stably reproduced across repeated experiments that realize different fiber elements.

The analogy with gauge theory is precise here. In a gauge theory, physical observables are those that are invariant under gauge transformations — the group acting on the fibers of the principal bundle. The algebra of gauge-invariant observables is the appropriate algebra of physical quantities. In the present framework, the equivalence relation $x \sim x'$ whenever $\pi(x) = \pi(x')$ plays the role of the gauge group, and $\mathcal{O}(M)$ plays the role of the gauge-invariant observable algebra. The admissible projection π is the quotient map of this gauge structure.

In the ARE setting, $\mathcal{O}(M)$ consists precisely of those functions of the aggregate transcript $\hat{y}$ that can be computed without access to the individual encodings. Simulation-based security is exactly the statement that any efficient computation on the evaluator's view belongs to $\mathcal{O}(M)$: the only efficiently computable function of the aggregate is the designated output $f(x_1, \dots, x_n)$.

In the bacterial setting, $\mathcal{O}(M)$ consists of those cellular observables that are independent of the exact microscopic trajectory ensemble: protein abundance, mRNA half-life, the coupling coefficient between transcription and translation. These are the quantities that remain stable under molecular stochasticity, spatial heterogeneity, and thermodynamic fluctuation. Quantities not in $\mathcal{O}(M)$ — such as the exact identity of which ribosome translates which transcript, or the precise spatial path of a given mRNA molecule — are not stably accessible to any measurement on the biological system operating at steady state.

The classical epistemological claim that the map is not the territory becomes, under this framework, a structural statement with precise content: the territory is X , the map is π , and the only quantities accessible to a system operating on the admissible manifold are those belonging to $\mathcal{O}(M)$. What the system *can* know is

determined not by the richness of X but by the structure of π . This is not merely epistemic incompleteness. The system itself only preserves the quotient geometry. The map is dynamically primary because the dynamical attractor *is* the projection class, not any element of its preimage.

The Symmetry Between Mixing and Separation

The two instances of topological privacy examined in this essay are related by a structural symmetry. Mixing enlarges fibers by aggregating many inputs into the same output. Separation eliminates trajectories by making certain regions of the state space geometrically unreachable. In both cases the observer is denied access to the microscopic interior. In mixing, the interior is crowded with indistinguishable alternatives. In separation, the interior is empty of reachable paths.

This symmetry suggests that topological privacy is not a single mechanism but a class of mechanisms unified by the common property that the fiber structure of the admissible projection is inaccessible to trajectories confined to the image. The specific realization — whether by entropy injection, spatial separation, temporal gating, or structural obstruction — is secondary to the abstract topological property it instantiates.

In the RSVP language, mixing corresponds to a large entropy field S in the image of π , while separation corresponds to a steep gradient in the admissibility field Φ that falls to zero across the boundary between admissible and inadmissible regions. Both configurations produce topological privacy. They are dual in the sense that each can be expressed as a limiting case of the other under appropriate deformations of the field triple $(\Phi, \mathbf{v}, S)$.

Part VI: Constraint, Entropy, and Persistent Computation

Constraint Before Content

The ARE and bacterial systems examined in this essay both exhibit a striking inversion of classical informational ontology. Traditionally, one imagines that content exists first, that computation manipulates that content, and that constraints are externally imposed afterward as a kind of friction. The systems studied here exhibit the opposite structure. The constraints determine which trajectories can exist before any stable functional content appears. The admissibility geometry precedes the invariant.

In the ARE setting, the evaluator cannot reconstruct provenance because the

shuffle and aggregation structure eliminate the relevant degrees of freedom before decoding occurs. The admissible geometry of the protocol determines what kinds of observables are even definable. No transcript produced by valid ARE encoders can carry individual input information; the structure of the encoding scheme makes such a transcript topologically impossible. Content — in the sense of function output — is the residue left after the constraint geometry has eliminated everything else.

In the bacterial setting, RNase accessibility, transcriptional coupling, and degradation kinetics are governed by spatial topology before any particular transcript fate is realized. The geometry determines which molecular trajectories can become stable. A transcript encodes the same genetic sequence regardless of whether it is coupled to a ribosome or prematurely released into the cytoplasm, but its fate — and therefore the functional content it can contribute to the cell — is determined entirely by geometry.

Remark. *This inversion is central to the RSVP framework. The stable object is not the microscopic configuration itself, but the persistence class induced by the admissible projection. Objects emerge from constrained flow. They are not fundamental primitives that later encounter constraints; they are the residue of constraint satisfaction.*

Persistent Computation Without Persistent Representation

One of the deepest implications of the ARE result is that universal computation does not require persistent local representation. The classical symbolic model of computation assumes stable memory registers, identifiable state transitions, recoverable intermediate representations, and explicit communication pathways. The shuffle/ARE framework undermines all four assumptions simultaneously.

The evaluator does not know which message came from which party. It cannot reconstruct intermediate representations. It observes only an aggregated projection. And yet it still computes arbitrary functions. This is not merely secure computation. It is computation surviving representational destruction.

The same phenomenon appears biologically. The cell does not preserve exact transcript trajectories, exact diffusion histories, exact ribosome-polymerase alignments, or exact degradation pathways. Nevertheless, functional regulation persists robustly across thousands of simultaneous transcription and translation events, across the stochastic variation inherent in molecular encounter dynamics, and across the spatial heterogeneity of the cellular interior.

Definition 8 (Persistent Computation). *A system performs persistent computation if it maintains stable functional invariants despite continual destruction, randomization, or inaccessibility of microscopic representational structure. Persistent computation is computation whose stability arises precisely because the invariant does not depend on microscopic recoverability.*

Persistent computation is therefore not computation with noise added afterward. It is computation organized from the outset around invariants that are indifferent to the microscopic paths by which they are reached. The admissible projection π is not an approximation; it is the exact structure of the computation.

Entropy as Functional Selector

Classical thermodynamic intuition often treats entropy as antagonistic to order and functionality. The systems studied here demonstrate a more nuanced picture. In ARE systems, increased entropy strengthens privacy and stabilizes the functional invariant by enlarging the fibers of π . Entropy destroys provenance while preserving computation. More randomness, injected locally, means larger fibers, which means stronger security and more robust invariant preservation. Entropy is here a resource, not a threat.

In weakly coupled bacterial systems, excessive entropy destabilizes the coupled transcriptional manifold by allowing trajectory divergence, Rho loading, and premature termination. The entropy that flows into the spacer between RNAP and ribosome is entropy that disrupts coupling, collapses the coupled trajectory, and sends the transcript into a high-degradation regime. Entropy here is a threat, not a resource.

Entropy therefore has no universal sign. Its effect on function depends entirely on the geometry of admissibility relative to the direction of entropic flow. The appropriate formulation is

$$\text{Persistence} \neq \text{low entropy.}$$

Rather,

$$\text{Persistence} = \text{stability of invariant structure under entropy flow.}$$

A system remains functional not by minimizing entropy universally, but by ensuring that entropy flow remains tangent to the admissible manifold M . When entropy flow exits the tangent bundle of M , the invariant destabilizes. In the ARE

setting, the injected entropy remains within the security-preserving quotient geometry; it enlarges fibers without perturbing the invariant. In the bacterial setting, low translation efficiency permits entropy flow into trajectory sectors that are transverse to the coupled manifold, thereby destabilizing the coupling that maintains transcript integrity.

The RSVP framework accommodates both behaviors naturally. The entropy field S is not intrinsically destructive or constructive. Its effect depends on the interaction between the scalar admissibility field Φ , the trajectory flow field $\mathbf{v}$, and the topology of the projection manifold M . When ∇S points into the tangent bundle of M , entropy increase is compatible with invariant stability. When ∇S points transversely to M , entropy increase destabilizes the projection.

Part VII: Toward a General Theory of Admissible Systems

Beyond Two Domains

The examples examined in this essay suggest that admissible projection is not a specialized mechanism restricted to cryptography or bacterial regulation. It may instead be a general organizational principle for complex systems in which functional stability must be maintained under conditions of partial irreversibility and continuous microscopic variation.

In morphogenesis, embryological development proceeds through massive irreversible projection. Cellular trajectories branch, signaling pathways diverge, epigenetic states become inaccessible, and yet stable anatomical invariants emerge with extraordinary reliability. The mature organism is not a preservation of microscopic developmental history. It is a stable projection class produced by constrained developmental flow under conditions of continuous entropy production. The full trajectory ensemble of every cell from zygote to adult is irretrievably lost; the organism is the invariant that survives this loss.

In cognition, perceptual systems discard vast quantities of sensory detail. Semantic systems collapse many distinct representations into invariant categories. Memory systems preserve compressed attractors rather than exact trajectories. The invariant that the cognitive system computes — the recognition of an object, the parsing of a sentence, the application of a concept — does not depend on exact reconstruction of the sensory or representational path. It is a property of the equivalence class, accessible through many distinct microscopic trajectories. The mind computes by projection, and its reliability arises from the same source as the reliability of ARE systems: the functional invariant is constant across large

fibers, and therefore robust to variation within them.

The Generalized Principle of Functional Persistence

The central principle emerging from this essay can now be stated in its general form.

Proposition 3 (Generalized Functional Persistence). *Let X be a microscopic trajectory space and let $\pi : X \rightarrow M$ be an admissible projection preserving a class of observables $\mathcal{F}$. A system exhibits persistent function if the observables in $\mathcal{F}$ are constant on sufficiently large fibers of π ; if entropy flow increases microscopic indistinguishability without destabilizing the admissible manifold M ; and if perturbations of microscopic trajectories remain confined to equivalence classes preserving $\mathcal{F}$.*

This principle unifies privacy-preserving computation, bacterial gene regulation, thermodynamic coarse-graining, semantic abstraction, and morphogenesis under a single geometric description. The common structure is not symbolic manipulation. It is admissible invariant persistence under irreversible projection. The principle does not depend on the nature of the microscopic substrate; it holds whenever an admissible projection exists with the stated properties.

The generalization also clarifies what distinguishes robust systems from fragile ones. A fragile system is one whose functional output depends on fiber-internal structure: small changes in the microscopic trajectory produce large changes in the observable. A robust system is one whose functional output is genuinely fiber-invariant: the same output is produced by a large and topologically connected class of microscopic realizations. Robustness, under this formulation, is not a property added on top of function. It is a consequence of the geometric relationship between the functional observable and the fiber structure of the admissible projection.

Admissible Projection and Renormalization

The structure of admissible projection is closely related to the logic of renormalization in theoretical physics, and making this connection explicit substantially broadens the scope of the framework. In the Wilsonian approach to renormalization, a quantum or statistical field theory is not described by a single action functional but by a family of effective theories parametrized by a cutoff scale Λ . Integrating out degrees of freedom below the scale Λ produces a coarser effective theory whose predictions for long-wavelength observables match those of

the original theory, while the short-wavelength microscopic structure becomes inaccessible.

The admissible projection $\pi : X \rightarrow M$ can be interpreted as a generalized renormalization operator. Introduce a scale parameter Λ and define a scale-dependent family of projections

$$\pi_\Lambda : X \longrightarrow M_\Lambda,$$

where M_Λ is the admissible manifold at scale Λ : the quotient of X by the equivalence relation that identifies all microscopic configurations indistinguishable by any observable of characteristic scale larger than Λ . As Λ decreases, M_Λ becomes coarser: fewer microscopic distinctions survive, and the fibers of π_Λ grow. As Λ increases, M_Λ becomes finer: more microscopic structure is retained, and the projection approaches the identity on X .

Under this formulation, persistent observables at scale Λ are precisely the elements of $\mathcal{O}(M_\Lambda)$: functions that are constant on the fibers of π_Λ . The flow from fine to coarse scale — decreasing Λ — is a flow in the space of projections, and the renormalization group equation governing this flow is structurally analogous to the RSVP admissibility conservation law. Renormalization fixed points correspond to admissible manifolds M_Λ that are self-similar under further coarsening: projections whose fiber structure does not change under small decreases in Λ . These are the most robust observables, stable not merely under a fixed projection but under the entire renormalization flow.

In the ARE setting, the relevant scale is the information scale: how much distinguishing power the evaluator has access to. At any finite information scale, the evaluator's accessible observables form a subalgebra of $\mathcal{O}(M)$. Security corresponds to the condition that the designated output $f(x_1, \dots, x_n)$ is the only element of $\mathcal{O}(M)$ accessible at the evaluator's scale. In the bacterial setting, the relevant scales are spatial and temporal: cellular observables at the scale of protein abundance and mRNA lifetime are renormalization fixed structures of the underlying molecular dynamics, stable precisely because the fiber structure of the projection from molecular trajectories to cellular observables is preserved under the biological analogue of scale flow.

Morphisms of Admissible Systems

The admissible projection framework admits a natural categorical formulation that makes precise what it means for two systems to be structurally equivalent, and what it means for one system to simulate another.

Define the category **Adm** whose objects are triples (X, π, M) consisting of a microscopic trajectory space X , an admissible manifold M , and an admissible projection $\pi : X \rightarrow M$. A morphism

$$F : (X, \pi_X, M_X) \longrightarrow (Y, \pi_Y, M_Y)$$

in **Adm** is a pair of maps $(F_X : X \rightarrow Y, \tilde{F} : M_X \rightarrow M_Y)$ such that the diagram

$$\begin{array}{ccc} X & \xrightarrow{F_X} & Y \\ \downarrow \pi_X & & \downarrow \pi_Y \\ M_X & \xrightarrow{\tilde{F}} & M_Y \end{array}$$

commutes: $\pi_Y \circ F_X = \tilde{F} \circ \pi_X$. The commutativity condition requires that F_X maps fiber elements of π_X to fiber elements of π_Y : if two microscopic configurations are equivalent under π_X , their images under F_X are equivalent under π_Y . Morphisms in **Adm** therefore preserve the admissible observable algebra: if $f \in \mathcal{O}(M_Y)$ then $f \circ \tilde{F} \in \mathcal{O}(M_X)$, meaning that any observation definable on the target system pulls back to a well-defined observation on the source system.

This categorical formulation makes precise several structural relationships that the essay has treated informally. The ARE scheme is a morphism from the trajectory space of joint inputs and randomness $(X_{\text{ARE}}, \pi_{\text{ARE}}, M_{\text{ARE}})$ to the output space, with the encoding map playing the role of F_X and the decoder playing the role of $\tilde{F}$. Security is the condition that this morphism has no nontrivial sections: there is no map from M_{ARE} back to X_{ARE} compatible with the categorical structure.

The biological regulatory system is a morphism from the space of molecular trajectory ensembles to the space of cellular observables. The membrane-localization constraint on RNase E is a condition on the morphism F_X : it restricts the image of F_X to trajectory ensembles in which degradation encounters are kinetically inaccessible, thereby ensuring that the induced map $\tilde{F}$ on the admissible manifold sends coupled transcriptional states to stable protein-production states.

Isomorphisms in **Adm** — morphisms with invertible $\tilde{F}$ — correspond to systems that preserve the same admissible invariant algebra through different microscopic mechanisms. The three bacterial species examined in Part III are isomorphic in **Adm** with respect to the functional observable of controlled gene expression, even though their microscopic implementations differ entirely. This categorical isomorphism is what the essay has called species-specific admissibility topology: different objects in **Adm** with the same admissible invariant structure.

Failure Modes: Catastrophic Projection Collapse

The essay has so far presented admissible projection primarily in its successful form: systems in which functional invariants survive microscopic collapse. An equally important part of the framework is the characterization of failure. Not all irreversible projections are admissible. When projection becomes too coarse, functional invariants are destroyed rather than preserved. This is what we call catastrophic projection collapse.

Formally, catastrophic collapse occurs when the projection π fails to respect the functional structure of the system. The precise condition is the existence of $x, x' \in X$ such that

$$f(x) \neq f(x') \quad \text{but} \quad \pi(x) = \pi(x').$$

That is, two microscopically distinct configurations that produce different functional outputs are mapped to the same point on the admissible manifold. The projection conflates functionally distinct states. This is not the benign fiber-enlargement of secure systems; it is a destructive coarsening that makes the invariant unrecoverable from the projected state.

In the ARE setting, catastrophic collapse corresponds to a decoding failure: two different function values produce indistinguishable aggregates. This is precisely what correctness requirements forbid. The ARE correctness condition

$$\Pr[\text{Dec}(\hat{y}) = f(x_1, \dots, x_n)] \geq 1 - \varepsilon$$

is the quantitative bound on how rarely catastrophic collapse is permitted to occur. Perfect correctness would require $\varepsilon = 0$, eliminating catastrophic collapse entirely; statistical correctness with negligible ε bounds its probability.

In the biological setting, catastrophic collapse corresponds to the failure of cellular regulation. When the admissibility constraint breaks down — when $\Gamma_{\text{RNase}} \cap \Gamma_{\text{nascent}}$ is no longer approximately empty, as in organisms with cytoplasmic RNase E or in experimental relocalization mutants — co-transcriptional degradation increases and the functional invariant of transcript stability becomes unreliable. The organism does not cease to function immediately, because the admissibility field Φ falls below $\Phi_{\min}$ only locally and transiently, but the robustness of the system against perturbation is reduced. A sufficiently severe disruption of the admissibility geometry would produce catastrophic collapse of the coupled transcriptional manifold.

The distinction between productive irreversibility and catastrophic collapse is therefore the central criterion for evaluating any admissible system. Productive

irreversibility enlarges fibers while preserving fiber invariance: the projection π maps more microscopic configurations to the same admissible output, but those configurations are functionally equivalent. Catastrophic collapse conflates functionally distinct configurations: the projection fails to separate f -preimages. The design of robust systems consists precisely in ensuring that entropy flow remains within the former regime and does not cross into the latter.

The deepest philosophical implication of the admissible projection framework is that microscopic ontology loses explanatory primacy. If stable function depends primarily on equivalence classes, admissibility geometry, and invariant-preserving projection, then the exact microscopic realization becomes secondary not merely epistemically but structurally.

This does not imply that microscopic structure is unreal. It implies that persistent systems are organized around projection classes rather than exact trajectories, and that the explanatorily relevant level of description is the level of the admissible manifold rather than the level of the microscopic configuration space. The ARE evaluator does not need the original messages; the bacterial cell does not preserve transcript histories; the cognitive system does not preserve raw sensory trajectories; the thermodynamic system does not preserve microscopic particle paths. What persists is the invariant.

Classical scientific ontology tends to privilege the microscopic as the locus of ultimate explanation. The tradition from Democritus through Boltzmann to molecular biology treats the macroscopic as supervening on the microscopic in a way that makes the macroscopic derivative and the microscopic foundational. The admissible projection framework inverts this priority not by denying the existence of microscopic structure but by demonstrating that it is the projection — the map $\pi : X \rightarrow M$ — and not the preimage that is explanatorily fundamental for any system whose outputs are persistent and computable.

The stability that matters for function is not the stability of trajectories but the stability of projection classes. The invariant that survives collapse is not located in any particular microscopic realization. It is located in the geometry of the map itself. Stability belongs not to the microscopic realization but to the projection class.

This observation can be consolidated as a general structural principle.

Proposition 4 (Projection Robustness Principle). *A dynamical system is robust precisely to the extent that its operational observables factor through an admissible projection $\pi : X \rightarrow M$ whose fibers absorb microscopic perturbations without altering invariant*

structure. That is, a system is robust with respect to a class of perturbations δx if and only if for all $x \in X$ and all admissible perturbations δx , $\pi(x + \delta x) = \pi(x)$: perturbations remain within the fiber and do not change the projection class.

This proposition unifies the robustness properties of the two systems examined in this essay. Shuffle MPC is robust under any permutation of messages, under any choice of randomness assignments by the local encoders, and under any substitution within the fiber $\pi^{-1}(\hat{y})$, because all such variations preserve the aggregate $\hat{y}$ and therefore leave $\pi(x)$ unchanged. Bacterial gene regulation is robust under molecular stochasticity, diffusion noise, and individual transcript fate variation, because these perturbations do not move the system off the admissible manifold M : the statistical observables, protein levels, mRNA half-lives, and coupling ratios, remain fiber-invariant under the relevant perturbation classes. Fragility enters precisely when a perturbation crosses a fiber boundary, moving the system to a different projection class and thereby altering the invariant structure. In the bacterial setting this corresponds to the transition from coupled to uncoupled transcription; in the ARE setting it would correspond to a break in the encoding structure that allows the evaluator to distinguish fiber elements.

Remark. *This is the sense in which function survives collapse. The collapse is not a failure of the system. It is the mechanism by which persistent invariants become possible. A system that preserved all microscopic detail would be a system in which every perturbation threatens every output. A system organized around admissible projection tolerates perturbation precisely because the invariant does not depend on the perturbed element. The collapse creates the robustness.*

Part VIII: Seed Development as Admissible Signalling Geometry

The Developing Seed as a System of Partially Isolated Dynamical Domains

A recent single-nucleus RNA sequencing atlas of early *Arabidopsis thaliana* seed development (Martin et al., 2026) provides a third major empirical instance of the admissible projection framework, one that extends its reach from computation and prokaryotic gene regulation into multicellular developmental biology. The study profiles 54,210 nuclei across 3, 5, and 7 days after pollination, capturing the transcriptional programs of the embryo, endosperm, and seed coat at high resolution. What the atlas reveals is not a centrally orchestrated developmental program but a distributed system of partially isolated signalling domains main-

taining developmental coherence across constrained interfaces.

The three main seed tissues are genetically distinct and symplastically separated: they cannot exchange arbitrary intracellular state through direct cytoplasmic continuity. Coordination therefore does not arise from unrestricted informational transparency but from sparse signalling across well-defined interfaces. This is the structural signature of an admissible system. The full microscopic biochemical state of each tissue is inaccessible to the others; what propagates across tissue boundaries is a low-dimensional projection sufficient to maintain global developmental invariants.

In the language of Part I, the developing seed instantiates an admissible projection

$$\pi : X_{\text{seed}} \longrightarrow M_{\text{seed}},$$

where X_{seed} is the full microscopic state space of all transcriptional, hormonal, and signalling dynamics across all tissues, and M_{seed} is the admissible developmental manifold: the space of tissue coordination states consistent with successful seed maturation. The critical observation is that $M_{\text{seed}} \neq \bigcup_i M_i$ for the individual tissue manifolds M_i . The global developmental outcome is not the union of isolated tissue programs; it emerges from their constrained interaction across admissibility interfaces.

Short Secreted Peptides as Low-Dimensional Coordination Operators

The atlas identifies the endosperm, particularly its micropylar and chalazal regions, as the primary hub for expression of short secreted peptides (SSPs), including members of the DEFL, LCR, and LTP families, as well as specifically characterized peptides including TWS1, ALE1, KRS, and RALFL3 (Martin et al., 2026). Many of these are upregulated after fertilization and are concentrated at the critical tissue interfaces where the symplastically isolated compartments must coordinate.

The SSPs are not carrying exhaustive developmental information across tissue boundaries. They are carrying constraint updates: low-dimensional signals sufficient to update the admissible developmental trajectory of the receiving tissue without requiring full informational transparency into the sending tissue's internal state.

This is the developmental analogue of the ARE encoding. A thermostat does not communicate the full microscopic thermodynamic state of a building. It communicates a single low-dimensional control signal. Likewise, the SSPs appear to

coordinate admissibility boundaries — thresholds of developmental readiness, timing of transitions, compatibility of nutrient-flow states — rather than transmitting full biochemical descriptions. The receiving tissue updates its trajectory on M_i in response to a low-dimensional signal from the interface, without accessing the internal state of the neighboring compartment.

This can be formalized as a coordination operator

$$\mathcal{C}_{ij} : M_i \longrightarrow M_j,$$

mapping the projected developmental state of tissue i to an update of the admissible state of tissue j . In the category **Adm** introduced in Part VII, these coordination operators are morphisms between the individual tissue admissible systems. The SSPs instantiate these morphisms physically: they are the molecular realization of low-bandwidth inter-manifold communication. Their sequence conservation across species, and the evidence for rapid evolution at tissue interfaces under conditions of genetic conflict, is precisely what one would predict of morphisms under selection pressure to maintain inter-manifold coordination while allowing microscopic variation within each tissue’s internal state.

The Chalazal Endosperm as a Developmental Interface Manifold

The chalazal endosperm (CZE) is the region of the endosperm closest to the vascular supply of the maternal tissue. The atlas reveals that the CZE is not a homogeneous compartment. It exhibits apical-basal transcriptional polarity within its coenocytic cyst: the gene *AT3G49307* marks the apical region while *RALFL3* marks the basal founder nuclei, which show transcriptional similarity to the adjacent maternal tissues, the chalazal proliferating tissue and the inner integument (Martin et al., 2026). Pseudotime analysis suggests the basal cyst follows an independent developmental trajectory from the rest of the CZE.

This result is of particular significance for the admissible projection framework because it demonstrates that even within a nominally single tissue compartment, signalling geometry remains anisotropic and directionally structured. The CZE is not merely a tissue. It is a developmental interface manifold, a geometric region where multiple admissibility flows converge, interact under directional constraints, and separate toward distinct developmental attractors.

In RSVP terms, the chalazal endosperm instantiates a non-trivial vector field $\mathbf{v}(x, t)$ within the tissue interior. Developmental trajectories are not isotropic diffusion but directed constrained transport along geometry-dependent admissi-

bility gradients. The apical-basal polarity is a manifestation of this directionality: the entropy field S and the admissibility density Φ have non-zero gradients within the cyst, producing distinct trajectory bundles in its apical and basal regions.

The basal founder nuclei, with their transcriptional similarity to adjacent maternal tissues, are especially interpretable through this lens. They occupy a region of the developmental manifold where the fiber structure of the projection π is shared with neighboring tissue compartments: the same admissible developmental states are realizable from more than one microscopic tissue identity. This is fiber overlap across tissue boundaries, and it is precisely what enables the chalazal region to function as an interface: it mediates between the endosperm's internal developmental trajectory and the maternal tissue's signalling geometry through a shared region of M_{seed} .

Brassinosteroid Localization and Curvature of Developmental Admissibility Space

One of the most spatially precise results in the atlas is the localization of brassinosteroid biosynthesis. The genes BR6OX1 and BR6OX2, encoding the final enzymes in the brassinosteroid synthesis pathway, are expressed with high specificity in a micropylar subtype of the outer integument (Martin et al., 2026). Brassinosteroid signalling, as reflected by expression of the BZR1/BES1 family transcription factors, is then detected in the micropylar endosperm at 5 days after pollination.

The spatial restriction of hormone production to a single specialized cell subtype represents a concrete biological realization of a non-uniform admissibility field. Brassinosteroid is not distributed homogeneously across the seed. It is produced at a localized source and establishes a spatial gradient in developmental admissibility space. In RSVP terms,

$$\nabla\Phi \neq 0$$

across the seed interior: the admissibility density Φ has a spatial structure determined by the geometry of the hormone source, not by centralized global instruction. Developmental behavior in neighboring tissues depends on position relative to this gradient rather than on access to any symbolic specification of what to do.

This is the developmental analogue of a gravitational or electromagnetic field. The brassinosteroid-producing micropylar outer integument behaves as a localized field generator whose influence propagates outward through concentration

gradients, inducing tissue-specific responses according to the local value of Φ rather than through explicit messaging. The receiving tissues in the micropylar endosperm do not need to know the internal state of the outer integument; they respond to the projected admissibility signal encoded in the local hormone concentration.

Development as Constraint Propagation Across Partially Isolated Manifolds

The atlas as a whole supports a reinterpretation of plant embryogenesis that replaces the implicit genetic blueprint metaphor with a constraint-propagation picture. The blueprint model implies that developmental outcomes are encoded in a centrally accessible program executed sequentially. The atlas evidence supports a different picture: global developmental coherence emerges from local constraint updates propagated across partially isolated tissue domains through sparse signalling interfaces.

No single tissue appears to contain or execute the complete developmental program for the seed. The embryo, endosperm, and seed coat each operate according to their own internal trajectories on their respective admissibility manifolds M_i . Global coordination emerges not from any tissue accessing the internal state of the others, but from the low-dimensional coordination signals — the SSPs, the hormonal gradients, the mechanical and nutrient flow cues — that update admissibility boundaries across interfaces.

The analogy to flocking behavior is exact. No bird contains the blueprint for the flock. Global structure emerges from local interaction constraints that each bird can compute from its immediate neighbourhood. Likewise, each seed tissue responds to locally accessible signals while the global seed developmental manifold emerges from their constrained interaction. The developmental system solves a distributed coordination problem without centralizing information, precisely as the ARE system solves a secure computation problem without centralizing state.

The rapid evolution observed at maternal-filial tissue interfaces, concentrated in extracellular domains and intrinsically disordered regions of secreted proteins (Martin et al., 2026), is the evolutionary fingerprint of this architecture. Interfaces under genetic conflict — where maternal and paternal interests in resource allocation diverge — are precisely the locations where the coordination operators $\mathcal{C}_{ij}$ are under the strongest selection pressure to change. The molecular machinery implementing inter-manifold communication evolves rapidly; the global developmental invariant it maintains evolves much more slowly. Function survives at

the level of M_{seed} while representation diverges at the level of the individual morphisms. This is the evolutionary expression of the Projection Robustness Principle: perturbations to the molecular implementation of coordination operators remain within fibers of the developmental projection, leaving the invariant structure of M_{seed} intact.

Part IX: The Formal Isomorphism Between Shuffle Privacy and Spatial Regulation

From Structural Analogy to Categorical Equivalence

The preceding parts of this essay have established that shuffle-model secure computation and spatial gene regulation in bacteria share the same abstract structure: both are instances of admissible projection $\pi : X \rightarrow M$ in which fiber-constant observables persist while microscopic provenance is destroyed. However, the claim of structural unity remains incomplete so long as the two systems are treated as analogous rather than formally equivalent. The purpose of this part is to close that gap by constructing a rigorous isomorphism, within the category **Adm**, between the shuffle operator governing ARE and the spatial accessibility operator governing RNase E–transcript interaction.

The strategy proceeds in five steps. First, the symmetry group of each system is identified and shown to act on the respective fiber bundle in structurally identical ways. Second, the projection operator for each system is characterized as a specific type of constraint Hamiltonian, and the two Hamiltonians are shown to satisfy the same structural equation. Third, the spatial collision integral formalizing the biological projection is derived and placed in correspondence with the aggregation functional of the ARE. Fourth, the obstruction cohomology class preventing admissible liftability is computed for both systems and shown to lie in the same cohomological position. Fifth, the invariant subalgebra lemma and the dynamical attractor theorem are proved in the general setting, completing the isomorphism.

Symmetry Groups and Fiber Bundle Structure

The first step in constructing the isomorphism is to identify the symmetry group of each system. In the ARE shuffle, the relevant symmetry group is the symmetric group S_n acting on the set of labeled inputs. Let X_{ARE} be the space of all n -tuples $(x_1, \dots, x_n, r_1, \dots, r_n)$ of inputs and randomness. The shuffle operator σ is a map

$\sigma : X_{\text{ARE}} \rightarrow M_{\text{ARE}}$ where $M_{\text{ARE}} = G$ is the Abelian group over which encodings are summed. The group S_n acts on X_{ARE} by permuting the labels:

$$\rho \cdot (x_1, \dots, x_n, r_1, \dots, r_n) = (x_{\rho(1)}, \dots, x_{\rho(n)}, r_{\rho(1)}, \dots, r_{\rho(n)}), \quad \rho \in S_n.$$

The aggregate sum $\hat{y} = \sum_i \text{Enc}(x_i; r_i)$ is manifestly permutation-invariant: $\sigma(\rho \cdot x) = \sigma(x)$ for all $\rho \in S_n$. The shuffle operator is therefore a quotient map for the S_n -action. The fiber $\sigma^{-1}(\hat{y})$ over any aggregate $\hat{y} \in G$ is the S_n -orbit of any element of X_{ARE} that maps to $\hat{y}$, together with all randomness assignments producing the same sum.

In the bacterial gene expression setting, the relevant symmetry group is the isometry group $\text{Iso}(3) = \mathbb{R}^3 \rtimes O(3)$ of Euclidean space, acting on the space X_{bio} of molecular trajectory ensembles. Let $\tau : X_{\text{bio}} \rightarrow M_{\text{bio}}$ be the spatial accessibility projection, where $M_{\text{bio}} = \{0, 1\}$ encodes the binary outcome of transcript stability. The membrane-targeting sequence of RNase E constrains the enzyme to a region within distance d_{min} of the inner membrane. Any isometry $g \in \text{Iso}(3)$ that preserves this distance constraint — that is, any g mapping the membrane-proximal region to itself — also preserves the accessibility partition of X_{bio} :

$$\tau(g \cdot \gamma) = \tau(\gamma) \quad \text{for all } \gamma \in X_{\text{bio}}, g \in \text{Stab}(\Omega_{\text{RNase}}),$$

where $\Omega_{\text{RNase}} \subset \mathbb{R}^3$ is the membrane-proximal region accessible to RNase E and $\text{Stab}(\Omega_{\text{RNase}})$ is the stabilizer subgroup of $\text{Iso}(3)$ preserving this region. The spatial projection τ is thus a quotient map for the action of this stabilizer.

Definition 9 (Structural Symmetry of an Admissible System). *An admissible system (X, π, M) has structural symmetry group G_π if G_π acts on X by a group action $\cdot : G_\pi \times X \rightarrow X$ such that $\pi(g \cdot x) = \pi(x)$ for all $g \in G_\pi$ and all $x \in X$. The fibers of π are then exactly the orbits of G_π .*

The ARE system has structural symmetry group $G_\sigma = S_n$. The bacterial system has structural symmetry group $G_\tau = \text{Stab}(\Omega_{\text{RNase}}) \leq \text{Iso}(3)$. These groups are not isomorphic as abstract groups, but they play the same structural role: each acts freely on the fibers of the respective projection, making the fiber over any base point $m \in M$ a torsor for the symmetry group. It is this structural role, not the group-theoretic identity, that defines the isomorphism.

Constraint Hamiltonians and the Projection Mechanism

The second step is to show that both projections arise from the same type of constraint Hamiltonian. A constraint Hamiltonian is a function $H_{\text{cons}} : X \rightarrow \mathbb{R}$ whose level sets define the admissibility partition. The projection π maps each $x \in X$ to the level set $\{x' : H_{\text{cons}}(x') = H_{\text{cons}}(x)\}$ quotiented by the symmetry group action.

For the ARE system, the constraint Hamiltonian is the additive aggregate:

$$H_{\text{cons}}^{\text{ARE}}(x_1, \dots, x_n, r_1, \dots, r_n) = \sum_{i=1}^n \text{Enc}(x_i; r_i) \in G.$$

The level sets of $H_{\text{cons}}^{\text{ARE}}$ are exactly the fibers of σ . The "collapse" of the ARE system consists in the evaluator accessing only the value of $H_{\text{cons}}^{\text{ARE}}$, not the individual terms in the sum. Since $H_{\text{cons}}^{\text{ARE}}$ is invariant under S_n , its level sets are unions of S_n -orbits, and the admissible observable algebra $\mathcal{O}(M_{\text{ARE}})$ consists exactly of functions of $H_{\text{cons}}^{\text{ARE}}$.

For the bacterial system, the constraint Hamiltonian is the encounter potential:

$$H_{\text{cons}}^{\text{bio}}(\gamma) = \int_0^T \mathbf{1}[\gamma(t) \in \Omega_{\text{RNase}}] dt,$$

where $\gamma : [0, T] \rightarrow \mathbb{R}^3$ is a trajectory of the nascent transcript and $\mathbf{1}[\gamma(t) \in \Omega_{\text{RNase}}]$ is the indicator that the transcript is within the RNase-accessible region at time t . When $H_{\text{cons}}^{\text{bio}}(\gamma) = 0$, the transcript has zero encounter probability and projects to the stable outcome; when $H_{\text{cons}}^{\text{bio}}(\gamma) > \theta$ for a threshold θ , the transcript projects to the degraded outcome. The level sets of $H_{\text{cons}}^{\text{bio}}$ are the fibers of τ , invariant under the stabilizer of Ω_{RNase} .

Both Hamiltonians have the form of a *constraint accumulator*: they sum or integrate a local quantity (the encoding contribution, or the accessibility indicator) over the relevant index set (parties, or time), and the admissible projection is the level-set map of this accumulator. The structural equation satisfied by both is:

$$\pi(x) = \pi(x') \iff H_{\text{cons}}(x) = H_{\text{cons}}(x'),$$

which is the defining condition for π to be the quotient map of the constraint Hamiltonian's level-set partition.

The Spatial Collision Integral

The biological constraint Hamiltonian $H_{\text{cons}}^{\text{bio}}$ can be made more precise through the theory of diffusion-limited reactions. The encounter probability between a nascent transcript at position $\mathbf{r}_0$ and RNase E distributed according to a density $\rho_{\text{RNase}}(\mathbf{r})$ on the inner membrane is given by the path-integral expression

$$P(\text{encounter} \mid \gamma) = 1 - \exp\left(-\kappa \int_0^T \rho_{\text{RNase}}(\gamma(t)) dt\right),$$

where κ is the intrinsic reaction rate and $\gamma(t)$ is the transcript's spatial trajectory. The expected encounter probability over the ensemble of transcript trajectories is

$$\bar{P}(\text{encounter}) = 1 - \mathbb{E}_\gamma \left[\exp\left(-\kappa \int_0^T \rho_{\text{RNase}}(\gamma(t)) dt\right) \right].$$

When RNase E is membrane-localized, $\rho_{\text{RNase}}(\mathbf{r})$ is concentrated on the membrane surface $\partial\Omega_{\text{cell}}$. For a transcript whose trajectory γ is confined to the nucleoid interior $\Omega_{\text{nucleoid}} \subset \Omega_{\text{cell}}$ at distance greater than d_{min} from the membrane, $\rho_{\text{RNase}}(\gamma(t)) \approx 0$ for all t , giving $\bar{P}(\text{encounter}) \approx 0$. This is the spatial admissibility constraint in quantitative form.

The spatial projection $\tau : X_{\text{bio}} \rightarrow M_{\text{bio}}$ is then defined by

$$\tau(\gamma) = \begin{cases} 0 & (\text{stable}) \quad \text{if } \bar{P}(\text{encounter} \mid \gamma) < \varepsilon, \\ 1 & (\text{degraded}) \quad \text{if } \bar{P}(\text{encounter} \mid \gamma) \geq \varepsilon, \end{cases}$$

for a threshold ε reflecting the minimum encounter probability required for degradation to proceed at a biologically significant rate. The fiber $\tau^{-1}(0)$ consists of all transcript trajectories for which the time-averaged density of RNase E along the path is below the degradation threshold.

Proposition 5 (Structural Equivalence of Constraint Accumulators). *Let $H_{\text{cons}}^{\text{ARE}}$ and $H_{\text{cons}}^{\text{bio}}$ be as defined above. Both satisfy the structural equation*

$$\pi(x) = \pi(x') \iff H_{\text{cons}}(x) = H_{\text{cons}}(x'),$$

and both are invariant under their respective structural symmetry groups. The admissible projection in each case is the level-set quotient map of the constraint accumulator.

Proof. For the ARE system: $H_{\text{cons}}^{\text{ARE}}$ is a group homomorphism from X_{ARE} (under componentwise encoding) to G . Its level sets are the fibers of σ . Invariance un-

der S_n follows from the commutativity of G : permuting the summands does not change the sum. The quotient map is σ by construction.

For the biological system: $H_{\text{cons}}^{\text{bio}}(\gamma) = \kappa \int_0^T \rho_{\text{RNase}}(\gamma(t)) dt$ is a functional of the trajectory γ . Its level sets define τ . Invariance under $\text{Stab}(\Omega_{\text{RNase}})$ follows from the fact that any isometry preserving Ω_{RNase} also preserves ρ_{RNase} pointwise, hence preserves the integral. The level-set map is τ by definition.

In both cases the projection is surjective onto M (every output value is achievable), the fibers are non-trivial (multiple elements of X map to each base point), and the symmetry group acts freely on fibers, making each fiber a torsor. The structural equations are identical in form. $\square$

The Invariant Subalgebra Lemma

Having established that both systems are governed by the same structural equation for their constraint accumulators, we now prove the invariant subalgebra lemma: any observable not invariant under the system's structural symmetry group is uncomputable by an observer restricted to M .

Proposition 6 (Invariant Subalgebra Lemma). *Let (X, π, M) be an admissible system with structural symmetry group G_π . An observable $f : X \rightarrow \mathbb{R}$ is computable by an observer restricted to M if and only if f is G_π -invariant, that is, $f(g \cdot x) = f(x)$ for all $g \in G_\pi$ and all $x \in X$.*

Proof. ($\Rightarrow$) Suppose f is computable from M , meaning there exists $\tilde{f} : M \rightarrow \mathbb{R}$ with $f = \tilde{f} \circ \pi$. For any $g \in G_\pi$ and $x \in X$:

$$f(g \cdot x) = \tilde{f}(\pi(g \cdot x)) = \tilde{f}(\pi(x)) = f(x),$$

where the second equality uses G_π -invariance of π . Hence f is G_π -invariant.

($\Leftarrow$) Suppose f is G_π -invariant. We must show f descends to M . Define $\tilde{f}(m) = f(x)$ for any $x \in \pi^{-1}(m)$. This is well-defined: if $x, x' \in \pi^{-1}(m)$ then $\pi(x) = \pi(x') = m$, so $x' = g \cdot x$ for some $g \in G_\pi$ (since the fibers are G_π -orbits), and G_π -invariance of f gives $f(x') = f(g \cdot x) = f(x)$. Thus $\tilde{f}$ is a well-defined function on M satisfying $f = \tilde{f} \circ \pi$. $\square$

Remark. For the ARE system, $G_\pi = S_n$ and the invariant observables are exactly the symmetric functions of the encodings, which by the ARE security condition reduce to functions of the aggregate sum alone. For the bacterial system, $G_\pi = \text{Stab}(\Omega_{\text{RNase}})$ and the invariant observables are exactly those depending only on the time-averaged encounter

probability — that is, on whether the transcript is stable or degraded, not on the specific trajectory by which it reached that state.

Obstruction to Admissible Liftability

We now prove the central structural theorem: neither system admits an admissible section, that is, a map $s : M \rightarrow X$ compatible with π that would allow an observer on M to recover the microscopic trajectory.

Definition 10 (Admissible Section). *A admissible section of $\pi : X \rightarrow M$ is a map $s : M \rightarrow X$ such that $\pi \circ s = \text{id}_M$ and s is compatible with the admissible observable algebra: for every $f \in \mathcal{O}(M)$, $f \circ \pi \circ s = f$.*

Proposition 7 (Obstruction to Liftability). *Let (X, π, M) be an admissible system with structural symmetry group G_π acting freely and non-trivially on the fibers of π . Then no admissible section $s : M \rightarrow X$ exists.*

Proof. Suppose for contradiction that an admissible section s exists. Fix any $m \in M$ and let $x_0 = s(m) \in \pi^{-1}(m)$. Since G_π acts freely and non-trivially on the fiber $\pi^{-1}(m)$, there exists $g \in G_\pi$, $g \neq e$, such that $g \cdot x_0 \neq x_0$. But $\pi(g \cdot x_0) = \pi(x_0) = m$, so both x_0 and $g \cdot x_0$ are elements of $\pi^{-1}(m)$.

Now consider the function $f : X \rightarrow \mathbb{R}$ defined by $f(x) = d(x, x_0)$ for some metric d on X . This function distinguishes x_0 from $g \cdot x_0$: $f(x_0) = 0$ while $f(g \cdot x_0) = d(g \cdot x_0, x_0) > 0$. However, f is not G_π -invariant (since $f(g \cdot x_0) \neq f(x_0)$), so by the invariant subalgebra lemma, $f \notin \mathcal{O}(M)$ and is uncomputable from M .

Any admissible section s would select a specific element $s(m) = x_0$ from the fiber $\pi^{-1}(m)$. But the selection criterion cannot be expressed in terms of observables from $\mathcal{O}(M)$ alone, since distinguishing x_0 from $g \cdot x_0$ requires a function outside $\mathcal{O}(M)$. Therefore no admissible selection rule exists, and s cannot be defined in a way compatible with the admissible observable algebra.

This is the obstruction. It is precisely the cohomological statement that the fibration $\pi : X \rightarrow M$ has no admissible section: the first Čech cohomology class $[\pi] \in \check{H}^1(M, \underline{G}_\pi)$ associated with the principal G_π -bundle is non-trivial, preventing the existence of a global section. $\square$

Remark. *For the ARE system, the non-triviality of $[\sigma] \in \check{H}^1(M_{\text{ARE}}, \underline{S}_n)$ is the formal statement that the shuffle cannot be inverted: knowing the aggregate $\hat{y}$ does not determine which of the $n!$ possible permutations of inputs produced it, and no admissible procedure can select a canonical representative of the fiber. For the bacterial system, the non-triviality*

of $[\tau] \in \check{H}^1(M_{\text{bio}}, G_\tau)$ is the formal statement that knowing a transcript is stable does not determine which of the many admissible nucleoid-interior trajectories it followed.

The central conclusion follows immediately.

Proposition 8 (Privacy and Regulation as Failure of Admissible Liftability).

$$\text{privacy/regulation} = \text{failure of admissible liftability}.$$

More precisely: the security of the ARE shuffle and the stability of the bacterial regulatory system are both equivalent to the non-existence of an admissible section of their respective projection maps, which is in turn equivalent to the non-triviality of the associated Čech cohomology class.

The Isomorphism Theorem

We can now state and prove the main theorem of this part: the ARE shuffle system and the bacterial spatial accessibility system are isomorphic objects in **Adm** with respect to their functional invariant structure, even though their structural symmetry groups are not group-theoretically isomorphic.

Definition 11 (Functional Isomorphism in **Adm**). *Two admissible systems (X_1, π_1, M_1) and (X_2, π_2, M_2) are functionally isomorphic if there exists an isomorphism $\tilde{F} : M_1 \rightarrow M_2$ in **Adm** such that the induced map on admissible observable algebras $\tilde{F}^* : \mathcal{O}(M_2) \rightarrow \mathcal{O}(M_1)$ is an algebra isomorphism, and both systems have non-trivial Čech cohomology obstruction classes preventing admissible liftability.*

Proposition 9 (Functional Isomorphism of ARE and Bacterial Systems). *The ARE admissible system $(X_{\text{ARE}}, \sigma, M_{\text{ARE}})$ and the bacterial spatial admissible system $(X_{\text{bio}}, \tau, M_{\text{bio}})$ are functionally isomorphic in **Adm** with respect to the following invariant structure: both have a designated computable functional observable (the function output $f(x_1, \dots, x_n)$ for ARE; the protein production rate for biology) that is the unique element of their respective admissible observable algebras $\mathcal{O}(M)$ that an observer on M can stably compute, and both have non-trivial obstruction classes preventing recovery of microscopic trajectory information.*

Proof. The proof proceeds by exhibiting the isomorphism $\tilde{F}$ explicitly and verifying its properties.

Define $\tilde{F} : M_{\text{ARE}} \rightarrow M_{\text{bio}}$ by mapping the subset of M_{ARE} corresponding to valid function outputs to the stable state $0 \in M_{\text{bio}}$, and mapping all other aggre-

gate values to the unstable state. More precisely, for the binary output case where $f(x_1, \dots, x_n) \in \{0, 1\}$: the aggregate $\hat{y}$ with $\text{Dec}(\hat{y}) = 0$ maps to $\tilde{F}(\hat{y}) = \text{stable}$, and $\hat{y}$ with $\text{Dec}(\hat{y}) = 1$ maps to $\tilde{F}(\hat{y}) = \text{degraded}$.

Under this map, the designated computable observable in ARE — the function output $f(x_1, \dots, x_n)$ — corresponds to the designated computable observable in the biological system — the binary stability state. The admissible observable algebras $\mathcal{O}(M_{\text{ARE}})$ and $\mathcal{O}(M_{\text{bio}})$ are both one-dimensional in the relevant subspace (one designated output observable, one binary state), and $\tilde{F}^*$ sends the stability observable to the function output observable, establishing the algebra isomorphism on the designated observables.

Both systems have non-trivial Čech obstruction classes by the Obstruction to Lifting proposition: the ARE shuffle has class $[\sigma] \in \check{H}^1(M_{\text{ARE}}, \underline{S}_n)$ and the bacterial system has class $[\tau] \in \check{H}^1(M_{\text{bio}}, \underline{G}_\tau)$, both non-trivial by the freeness and non-triviality of the respective group actions on fibers.

The functional isomorphism is therefore established: both systems preserve exactly the same abstract structure (one designated computable observable, non-trivial lifting obstruction, fiber-invariant algebra), through different physical mechanisms (permutation symmetry and additive aggregation for ARE; spatial symmetry and diffusion geometry for biology) that play the same structural role in **Adm**. $\square$

The Dynamical Attractor Theorem

The final component of the formal isomorphism is the convergence theorem: under the RSVP field dynamics, all trajectories in X satisfying the admissibility constraints converge to M . This gives the isomorphism its dynamical rather than merely static content.

Proposition 10 (Dynamical Attractor Theorem). *Let $(\Phi, \mathbf{v}, S)$ be the RSVP field triple on a domain $\mathcal{D}$ with Lagrangian density $\mathcal{L} = \frac{1}{2}|\mathbf{v}|^2 - V(\Phi) - \alpha S + \beta \Phi \nabla \cdot \mathbf{v}$. Suppose the constraint potential $V(\Phi)$ is strictly convex on $\{\Phi \geq \Phi_{\min}\}$ with a unique minimum on the admissible manifold M . Then under the Euler–Lagrange flow of $\mathcal{L}$ with entropy non-decrease, every trajectory $\phi_t(x)$ with $\Phi(\phi_t(x)) \geq \Phi_{\min}$ converges to M as $t \rightarrow \infty$.*

Proof. The Euler–Lagrange equation for $\mathbf{v}$ from $\mathcal{L}$ gives

$$\frac{D\mathbf{v}}{Dt} = -\nabla V(\Phi) + \beta \nabla(\Phi \nabla \cdot \mathbf{v}),$$

where D/Dt is the material derivative along the flow. The admissibility conservation equation $\partial_t \Phi + \nabla \cdot (\Phi \mathbf{v}) = -\lambda S$ implies that Φ decreases along trajectories proportionally to local entropy production.

Define the Lyapunov function $\mathcal{V}(x) = V(\Phi(x)) + \alpha S(x)$. Along any trajectory $\phi_t(x)$:

$$\frac{d}{dt} \mathcal{V}(\phi_t(x)) = \nabla V(\Phi) \cdot \partial_t \Phi + \alpha \partial_t S.$$

From the admissibility conservation equation, $\partial_t \Phi = -\nabla \cdot (\Phi \mathbf{v}) - \lambda S \leq -\lambda S$ (since $\nabla \cdot (\Phi \mathbf{v}) \geq 0$ in the admissible regime by the flow structure). The entropy non-decrease condition gives $\partial_t S \geq 0$.

For Φ near but above $\Phi_{\min}$, strict convexity of V on the admissible region implies $\nabla V(\Phi) \cdot (-\lambda S) \leq 0$ with equality only on M (where V achieves its minimum). Combined with the entropy non-decrease, $\mathcal{V}$ is a decreasing function along trajectories except on M . By the LaSalle invariance principle applied to the Euler–Lagrange flow, all trajectories in the admissible region $\{\Phi \geq \Phi_{\min}\}$ converge to the largest invariant subset of $\{\frac{d}{dt} \mathcal{V} = 0\}$, which by strict convexity is exactly M . $\square$

Remark. *This theorem applies to both the ARE and bacterial systems under their respective RSVP interpretations. For ARE, the Lyapunov function $\mathcal{V}$ measures the deviation of the encoding distribution from the security-preserving quotient geometry; the Euler–Lagrange flow drives the encoding distribution toward the maximal-entropy fiber structure on M_{ARE} . For the bacterial system, $\mathcal{V}$ measures the deviation of the transcript trajectory ensemble from the membrane-separated admissible region; the RSVP flow drives transcript trajectories toward the stable nucleoid-interior attractor basin on M_{bio} .*

Synthesis: The Unified Structural Picture

The results of this part can now be assembled into a single unified structural picture. Both the ARE shuffle system and the bacterial spatial regulation system are admissible systems in **Adm** with the following common properties, established by the preceding propositions and theorems.

Their projections arise from constraint Hamiltonians of the same structural type: they are both constraint accumulators, additive over an index set, invariant under a structural symmetry group acting freely on fibers. Their admissible observable algebras consist exactly of the invariant subalgebras of their respective symmetry groups. No admissible section exists for either: both carry non-trivial Čech cohomology obstruction classes that prevent microscopic recov-

ery from manifold-level observation. Both are dynamical attractors under RSVP field evolution: the Lyapunov function $\mathcal{V}$ decreases to zero on M under the constrained Euler–Lagrange flow. They are functionally isomorphic in **Adm**: there exists an isomorphism of their designated observable structures that respects the admissible algebra and the cohomological obstruction.

The formal conclusion is that privacy in shuffle-model computation and spatial regulation in bacterial gene expression are not merely analogous but are isomorphic instances of the same abstract phenomenon: admissible projection with non-trivial liftability obstruction, preserving a designated computable invariant while making microscopic provenance unrecoverable. The physical mechanisms differ entirely, but the structural position of those mechanisms within the category **Adm** is identical.

Interlude: The Three-Layer Architecture and Bubble Boundaries

Separating the Foundational, Dynamical, and Realization Layers

The material developed in Parts I through IX operates simultaneously across three distinct levels of abstraction, and the coherence of the monograph as a whole benefits from making these levels explicit before proceeding. The three layers are: the foundational geometric layer, the dynamical layer, and the domain realization layer. Each layer has its own vocabulary, its own formal objects, and its own mode of argument; the power of the framework derives from the fact that all three are structurally consistent instantiations of the same underlying geometry.

The *foundational geometric layer* consists of the objects and relations that require no dynamical or empirical specification to be defined. Its primary components are the admissible projection $\pi : X \rightarrow M$, the admissible invariant manifold M , the admissible observable algebra $\mathcal{O}(M)$, the structural symmetry group G_π , the obstruction class $[\pi] \in \check{H}^1(M, \underline{G}_\pi)$, and the bubble boundary ∂B defined below. These objects are axiomatic in the sense that they can be specified independently of any particular physical or computational realization.

The *dynamical layer* consists of the RSVP field triple $(\Phi, \mathbf{v}, S)$ and the associated Lagrangian $\mathcal{L} = \frac{1}{2}|\mathbf{v}|^2 - V(\Phi) - \alpha S + \beta \Phi \nabla \cdot \mathbf{v}$, its Euler–Lagrange flow, and the collapse operators that govern how systems move through admissibility space. This layer explains the temporal evolution of admissible systems: it answers the question of how a system approaches M , how it remains on M under perturbation, and how it exits M under catastrophic collapse. The RSVP framework is the dynamical layer. The Spheredrop calculus governs discrete collapse events

and irreversible quotient formation within this layer; the Yarncrawler framework governs the transport geometry connecting distinct admissibility basins.

The *domain realization layer* consists of specific instantiations of the foundational and dynamical structures in empirical or computational systems. Each domain realization names concrete objects in the abstract framework: for ARE, X is the input-randomness space, π is the shuffle operator, and $G_\pi = S_n$; for bacterial gene regulation, X is the molecular trajectory ensemble, π is the spatial accessibility projection, and $G_\pi = \text{Stab}(\Omega_{\text{RNase}})$; for seed development, X is the space of tissue-level biochemical states, π is the inter-tissue coordination projection, and the coordination operators $\mathcal{C}_{ij}$ are the morphisms in **Adm**. Additional domain realizations will be developed in Parts X and beyond, including typography, semantic cognition, and generative computation.

The relationship between layers is: the foundational layer provides the static categorical scaffolding; the dynamical layer provides the flow equations governing how systems traverse that scaffolding; and the realization layer interprets specific physical or computational systems within it. A theorem proved at the foundational layer holds in all domain realizations. A result derived at the dynamical layer applies wherever the RSVP field equations govern the relevant flows. A claim made at the realization layer holds for that specific domain and may or may not generalize.

Bubble Boundaries and the Geometry of Collapse

The most important geometric object not yet given an explicit definition in the main text is the bubble boundary ∂B . This object formalizes the intuition that admissible systems occupy a region of trajectory space with a definite boundary, crossing which constitutes collapse.

Definition 12 (Admissibility Bubble). *Let (X, π, M) be an admissible system with structural symmetry group G_π . The admissibility bubble $B \subseteq X$ is the maximal open set of configurations that project to admissible outputs under small perturbations:*

$$B = \{x \in X \mid \exists \varepsilon > 0 : \pi(x + \delta x) = \pi(x) \text{ for all } \|\delta x\| < \varepsilon\}.$$

The bubble boundary is

$$\partial B = \{x \in X \mid \forall \varepsilon > 0, \exists \delta x \text{ with } \|\delta x\| < \varepsilon \text{ and } \pi(x + \delta x) \neq \pi(x)\}.$$

Elements of B are robust under infinitesimal perturbation: small changes in

the microscopic trajectory do not alter the projection class. Elements of ∂B are marginally stable: arbitrarily small perturbations can change the projection class. Elements of $X \setminus (B \cup \partial B)$ are already outside the admissible regime.

The bubble boundary is the geometric locus of incipient collapse. In the Projection Robustness Principle proved in Part VII, the condition $\pi(x + \delta x) = \pi(x)$ for all admissible perturbations is precisely the condition that $x \in B$. Fragility is the condition $x \in \partial B$. Catastrophic projection collapse, as defined in the failure modes subsection, is the transition $x \rightarrow x + \delta x$ with $x + \delta x \notin B \cup \partial B$.

Proposition 11 (Bubble Boundary as Entropy Tangency Threshold). *Under the RSVP field dynamics, $x \in \partial B$ if and only if $\nabla S(x)$ acquires a non-zero component transverse to the tangent bundle of M at $\pi(x)$:*

$$x \in \partial B \iff (\nabla S(x))_{\perp} \neq 0,$$

where $(\nabla S)_{\perp}$ denotes the component of ∇S in the normal bundle $N(M) = TX|_M/TM$.

Proof. By the RSVP Lagrangian, the admissibility density Φ decreases at rate λS along trajectories. The condition $x \in B$ requires that this decrease is tangential to M : the projection class $\pi(x)$ remains constant under the flow. This tangentiality condition is $\partial_t \Phi|_M = -\lambda S|_M$ remaining in TM , which holds if and only if ∇S has no transverse component to M . When $(\nabla S)_{\perp} \neq 0$, the entropy gradient drives the admissibility density off M , causing the projection class to change, placing x at or beyond ∂B . $\square$

This proposition is the precise form of the entropy tangency condition mentioned throughout the essay. Entropy is destructive not in itself but in its direction: entropy flow tangent to M enlarges fibers without altering projection classes; entropy flow transverse to M drives the system toward ∂B and into collapse.

The bubble boundary formalism subsumes many phenomena discussed in earlier parts. Premature transcription termination in bacteria is the crossing of ∂B by the RNAP–ribosome coupled trajectory, driven by transverse entropy flow from the spacer RNA into the Rho-loading regime. Privacy leakage in ARE is the crossing of ∂B by an encoding distribution, allowing an adversary to distinguish fiber elements. Developmental failure in the seed system is the collapse of coordination operators $\mathcal{C}_{ij}$ when tissue interface geometry places the signalling system at ∂B . Each of these failure modes is a boundary-crossing event in the same geometric sense.

The Four-Layer Stack: RSVP, Spherepop, Yarncrawler, and **Adm**

The full theoretical architecture of this monograph involves four interacting frameworks that operate at complementary levels of description. Making their relationships precise is necessary for the monograph’s coherence as it expands into new domains.

RSVP provides the *continuous field dynamics*: the equations governing how the triple $(\Phi, \mathbf{v}, S)$ evolves over time and space, how admissibility density is transported and dissipated, and how the constrained Euler–Lagrange flow drives trajectories toward attractor basins on M . RSVP is the physics of the dynamical layer.

Spherepop provides the *collapse topology*: the discrete event structure governing irreversible quotient formation, bubble stabilization, and the moment of projection collapse. Where RSVP tracks the continuous approach to ∂B , Spherepop governs what happens at and after the crossing: the Pop operator collapses a bubble into its projection class, the Bind operator establishes a new equivalence relation, the Collapse operator finalizes the quotient, and the Refuse operator prevents certain transitions. Spherepop is the topology of irreversible events within the dynamical layer.

Yarncrawler provides the *inter-bubble transport geometry*: the mechanisms by which information, constraint updates, or coordination signals are transmitted between distinct admissibility basins. Where RSVP governs motion within a single admissibility regime and Spherepop governs the transition between regimes, Yarncrawler governs the network structure connecting multiple regimes. In the seed development context, Yarncrawler corresponds to the signalling channels carrying SSP coordination operators between tissue compartments. In the cognitive context, it corresponds to the associative pathways connecting distinct semantic attractor basins.

The category **Adm** provides the *categorical abstraction layer*: the language in which all of the above can be stated domain-independently. Objects in **Adm** are tuples $(X, \pi, M, \mathcal{O})$; morphisms are commutative diagrams preserving observable algebras; functors between **Adm** and other categories formalize the relationships between different domain realizations; natural transformations between such functors capture the structural equivalences that the isomorphism theorems of Part IX make precise.

The four layers are not competing accounts but complementary ones: RSVP provides the differential geometry, Spherepop provides the combinatorial topology, Yarncrawler provides the graph-theoretic transport structure, and **Adm** provides the categorical abstraction that lets results proved in any one domain trans-

fer to all structurally isomorphic ones.

Part X: Typography as Admissible Projection Geometry

Glyphs as Invariant Manifolds

The theory of admissible projection, developed in abstract form in Parts I through IX, finds one of its most immediately visualizable instantiations in typography. A glyph is not a bitmap. It is not a Bézier curve specification. It is not a set of stroke parameters. A glyph is an equivalence class: the set of all possible geometric realizations that an observer would recognize as the same symbolic identity. The letter A, in any font, in any weight, in any orientation permitting recognition, in any rendering medium from ink to pixel to chalk dust, is the same element of a semantic projection manifold. The glyph is the projection class, not any of its realizations.

This observation is not merely a metaphor for the abstract framework. It is a direct instance of it. Define the glyph trajectory space X_{stroke} to be the high-dimensional space of all possible geometric realizations of a single glyph: all possible stroke trajectories, curvature profiles, weight distributions, spatial frequency components, and rendering parameters. Define the semantic glyph manifold M_{semantic} to be the low-dimensional space of recognizable symbolic identities. The glyph projection is

$$\pi_{\text{glyph}} : X_{\text{stroke}} \longrightarrow M_{\text{semantic}},$$

mapping each geometric realization to its equivalence class under perceptual recognition. The fiber $\pi_{\text{glyph}}^{-1}(A)$ over the symbol A consists of all geometric realizations that an observer projects onto the symbolic identity A : Roman, bold, italic, condensed, extended, handwritten, graffitied, engraved, rendered in fog, approximated in ASCII — these are all elements of the same fiber.

The admissible observable algebra $\mathcal{O}(M_{\text{semantic}})$ consists of exactly those functions of glyph geometry that depend only on the projection class: functions that are constant on fibers of π_{glyph} . Semantic identity is in $\mathcal{O}(M_{\text{semantic}})$; exact stroke curvature at a specific point is not. This is precisely the distinction that makes reading possible: human perceptual systems compute elements of $\mathcal{O}(M_{\text{semantic}})$ without accessing fiber-internal detail.

Definition 13 (Glyph Admissibility Bubble). *The admissibility bubble $B_g \subset X_{\text{stroke}}$*

of a glyph $g \in M_{\text{semantic}}$ is the maximal open set of geometric realizations $x \in X_{\text{stroke}}$ such that

$$\pi_{\text{glyph}}(x + \delta x) = g \quad \text{for all sufficiently small perturbations } \delta x.$$

The bubble boundary ∂B_g is the set of marginal realizations at which arbitrarily small perturbations may change the recognized symbol.

The recognizability functional is defined as

$$R(x) = \Pr[\pi_{\text{glyph}}(x) = \text{intended glyph}],$$

where the probability is over the relevant ensemble of observers and perceptual conditions. Then $B_g = \{x : R(x) > R_{\min}\}$ for a threshold $R_{\min}$, and $\partial B_g = \{x : R(x) = R_{\min}\}$. A realization x with $R(x)$ well above $R_{\min}$ is deeply within the admissibility bubble: it is robust to substantial geometric perturbation. A realization with $R(x)$ near $R_{\min}$ is at the bubble boundary: marginal perturbations may collapse it to a different projection class or to no recognized symbol at all.

Semantic Liftability Under Geometric Distortion

The obstruction to admissible liftability proved in Part IX applies directly to the glyph system. An observer restricted to M_{semantic} — an observer who knows only the recognized symbolic identity, not the geometric realization — cannot reconstruct the specific element of X_{stroke} that produced it. This is not an empirical limitation but a structural one. Many distinct geometric realizations produce the same recognized symbol, and without additional information outside $\mathcal{O}(M_{\text{semantic}})$, no selection criterion distinguishes them.

The liftability obstruction for the glyph system can be stated as follows. The structural symmetry group $G_{\pi_{\text{glyph}}}$ of the glyph projection consists of all geometric transformations of X_{stroke} that preserve the recognized identity: stroke-width rescaling, compatible weight changes, admissible curvature deformations, and rendering-medium variations that leave π_{glyph} invariant. This group acts freely on the fibers of π_{glyph} , and the associated Čech cohomology class $[\pi_{\text{glyph}}] \in \check{H}^1(M_{\text{semantic}}, \underline{G_{\pi_{\text{glyph}}}})$ is non-trivial, preventing any admissible section.

This non-triviality is the formal statement that no canonical font exists: there is no privileged geometric realization of any glyph that could serve as the unique representative of its projection class without making an arbitrary choice of fiber element.

The RSVP Dynamics of Font Transformation

Font transformations — from regular to bold, from upright to italic, from formal to frenetic, from legible to hypnagogic — are flows on X_{stroke} parameterized by the RSVP field triple $(\Phi, \mathbf{v}, S)$. In this context:

The scalar field $\Phi : X_{\text{stroke}} \rightarrow \mathbb{R}$ measures the *recognizability density*: $\Phi(x)$ is high when x is deep within the admissibility bubble B_g and decreases toward zero as x approaches ∂B_g . High Φ indicates a realization far from collapse; low Φ indicates proximity to the bubble boundary.

The vector field $\mathbf{v} : X_{\text{stroke}} \rightarrow TX_{\text{stroke}}$ governs the *stylistic deformation flow*: the direction and rate at which a geometric realization moves through the trajectory space under a given typographic transformation. Bold transformation is a flow in the direction of increasing stroke mass. Italic transformation is a directional shear flow. Condensation is a compressive flow in the horizontal direction. Each typographic operation defines a specific vector field on X_{stroke} .

The entropy field $S : X_{\text{stroke}} \rightarrow \mathbb{R}$ measures the *local geometric disorder* of the realization: high S corresponds to high local variability in stroke geometry, many competing curvature interpretations, or high spatial frequency noise. Low S corresponds to clean, unambiguous geometric specification.

The admissibility conservation equation $\partial_t \Phi + \nabla \cdot (\Phi \mathbf{v}) = -\lambda S$ then expresses the constraint that recognizability density is transported by the stylistic flow $\mathbf{v}$ and dissipated at a rate proportional to local geometric entropy. A typographic transformation that increases S near the bubble boundary is a transformation that threatens recognizability.

Bold and Italic as Low-Entropy Manifold Flows

Bold and italic are the canonical examples of *admissible low-entropy flows*: transformations that move a glyph realization through X_{stroke} while remaining well within the admissibility bubble B_g . They increase or redirect geometric mass without crossing the bubble boundary.

A bold transformation $\phi_{\text{bold}} : X_{\text{stroke}} \rightarrow X_{\text{stroke}}$ expands the stroke width uniformly, increasing the admissibility density Φ in the sense of making the realization more visually prominent and reducing the probability of misrecognition due to small-scale rendering artifacts. Formally, bold transformation is a flow in the direction $\nabla \Phi_{\text{mass}}$, the gradient of the stroke-mass density component of Φ . The entropy field S changes minimally under bold transformation in the interior of B_g : the geometric structure of the glyph is preserved, only its weight is increased.

The admissibility conservation equation is satisfied with λS remaining small.

An italic transformation $\phi_{\text{italic}} : x \mapsto Ax$ is a shear map $A : X_{\text{stroke}} \rightarrow X_{\text{stroke}}$ with $A = I + \theta E_{12}$ for a shear angle θ and the elementary matrix E_{12} acting on the horizontal displacement of vertical strokes. The shear introduces a directional vector flow: $\mathbf{v}_{\text{italic}} = \theta \partial_x$, a constant horizontal displacement field. For small θ (typical typographic italic angles are 7–15 degrees), the flow remains within B_g : the shear is admissible because horizontal displacement preserves the topological skeleton of the glyph and the perceptual projection π_{glyph} is invariant under these shears.

Both bold and italic are examples of fiber-preserving flows in the sense of the Projection Robustness Principle: $\pi_{\text{glyph}}(\phi(x)) = \pi_{\text{glyph}}(x)$ for all x in the interior of B_g .

Frenetic Typography and Tangential Entropy Injection

Frenetic typography represents a qualitatively different flow regime: one in which the entropy field S is substantially elevated, but the entropy flow remains predominantly tangential to M_{semantic} . A frenetic realization of a glyph introduces high local variability — jagged edges, varying stroke weights, disrupted continuity, irregular spacing — while preserving enough of the topological skeleton that the projection π_{glyph} remains defined and stable.

In RSVP terms, frenetic typography increases S substantially but keeps $(\nabla S)_\perp \approx 0$: the entropy gradient is approximately tangential to M_{semantic} at the projected point. By the Bubble Boundary as Entropy Tangency Threshold proposition, the realization therefore remains within B_g despite elevated entropy. The recognizability density Φ decreases under the entropy injection — the realization is less far from the bubble boundary than a clean regular rendering — but does not reach zero.

This explains the phenomenology of frenetic typography: it reads as energetic, turbulent, or chaotic, because the elevated S is perceptible as visual noise, but it remains readable because the projection π_{glyph} is still well-defined. The fiber of the projection has expanded under entropy injection (more diverse geometric realizations map to the same glyph), making the system less fragile in some directions while also reducing the distance to ∂B_g .

Formally, define the frenetic flow as

$$\phi_{\text{fren}} : x \mapsto x + \xi(x),$$

where $\xi : X_{\text{stroke}} \rightarrow X_{\text{stroke}}$ is a high-frequency perturbation field satisfying $\mathbf{E}[\xi] = 0$ (zero mean) and $\text{Var}[\xi]$ large but confined to directions tangential to M_{semantic} . The condition $\text{Var}[\xi]_{\perp} \approx 0$ is the entropy tangency condition; it is the mathematical content of "frenetic but readable."

Hypnagogic Typography and Boundary Oscillation

Hypnagogic typography represents the most geometrically delicate regime: that of slow oscillation near the bubble boundary ∂B_g . A hypnagogic realization does not inject high-frequency entropy (as frenetic does) but instead introduces low-frequency drift in the admissibility field, weakening the local curvature constraints that define the glyph's geometric identity and allowing the projection class to become momentarily ambiguous.

In RSVP terms, hypnagogic typography produces a slowly varying admissibility field $\Phi(x, t)$ that oscillates near $\Phi_{\min}$: the system spends time near ∂B_g without permanently crossing it. The entropy field S is not elevated in total, but its transverse component $(\nabla S)_{\perp}$ periodically becomes non-zero before relaxing back to zero. The realization oscillates between stable projection (within B_g) and marginal projection (on ∂B_g), never committing fully to catastrophic collapse.

This explains the perceptual phenomenology of hypnagogic typography: it feels dreamlike or liminal because the observer's perceptual projection π_{glyph} is periodically destabilized. The glyph is recognized, then partially dissolved, then re-recognized, in a cycle governed by the slow oscillation of $\Phi(x, t)$ around $\Phi_{\min}$. Unlike frenetic typography, which is turbulent but stable, hypnagogic typography is smooth but marginally stable. Unlike catastrophic collapse, which permanently exits B_g , hypnagogic typography returns to B_g after each excursion.

The formal model is a trajectory $x(t) \in X_{\text{stroke}}$ satisfying the damped oscillation

$$\ddot{\Phi}(x(t)) + \gamma \dot{\Phi}(x(t)) + \omega_0^2 (\Phi(x(t)) - \Phi_{\min}) = \eta(t),$$

where γ is a damping coefficient, ω_0 is the natural oscillation frequency of the admissibility field around $\Phi_{\min}$, and $\eta(t)$ is a low-frequency noise term representing the slow geometric drift. The realization spends most of its time near $\Phi_{\min}$, occasionally dipping below (brief collapse, perceptual ambiguity) and recovering (return to recognition).

Recognizability Thresholds and Catastrophic Collapse

Catastrophic collapse in typography is the crossing of ∂B_g without return: a deformation so large that $\pi_{\text{glyph}}(x)$ no longer maps to the intended glyph, and the observer either recognizes a different symbol or recognizes nothing. This is the typographic instance of the failure mode formalized in Part VII.

The condition for catastrophic collapse is

$$\exists \delta x : \pi_{\text{glyph}}(x + \delta x) \neq g \quad \text{and} \quad \pi_{\text{glyph}}(x + \delta x) \notin B_g.$$

The realization has not merely moved to a different fiber of the same glyph but has exited the admissibility bubble entirely, entering either the fiber of a different glyph (a misrecognition) or the complement of all admissible regions (illegibility).

Recognizability as a function of deformation strength follows a characteristic profile. For small deformations within B_g , $R(x + \delta x) \approx R(x) > R_{\min}$: the glyph is robustly recognized. As the deformation approaches ∂B_g , $R(x + \delta x)$ decreases toward $R_{\min}$: recognition becomes uncertain. At ∂B_g , $R(x) = R_{\min}$: recognition is marginal. Beyond ∂B_g , R drops sharply as the perceptual system's projection π_{glyph} either misidentifies the symbol or fails entirely.

The transition from marginal to catastrophic is a phase transition in the RSVP sense: the Lyapunov function $\mathcal{V}(x) = V(\Phi(x)) + \alpha S(x)$ passes through a critical point at ∂B_g , and the trajectory exits the basin of attraction of M_{semantic} at g . The sharp drop in R at the boundary corresponds to the exit from the attractor basin proved in the Dynamical Attractor Theorem.

Reading as Projection Recovery

The typographic section makes a claim that goes beyond typography: reading itself is a form of admissible projection recovery. An observer reading a text does not reconstruct the exact geometric parameters of each glyph. The observer computes $\pi_{\text{glyph}}(x)$ from noisy, partial, perspective-distorted, illumination-variable sensory data x , extracting the projection class without reconstructing the preimage.

This is the dynamical attractor theorem applied to perceptual cognition. The observer's neural computation is an approximation to the RSVP flow that drives trajectory representations toward the attractor basin of M_{semantic} . The perceptual system is robust to optical distortion, font variation, partial occlusion, and rendering noise because all of these are perturbations within the admissibility bubble:

they change x while preserving $\pi_{\text{glyph}}(x)$.

Dyslexia, under this framework, is a condition in which the effective ∂B_g for certain glyph pairs is pathologically narrow: the admissibility bubbles of visually similar symbols overlap or are insufficiently separated, making the projection π_{glyph} unstable near the boundaries between similar glyphs such as b/d or p/q. The observer’s perceptual system cannot reliably determine which basin of attraction the sensory input falls into.

Adversarial typography — the deliberate construction of glyphs that fool optical character recognition systems while remaining human-readable — is the construction of realizations x near $\partial B_g^{\text{human}}$ but far inside B_g^{OCR} for a different glyph, exploiting the fact that human and machine perceptual systems define different admissibility bubbles over the same trajectory space. The adversarial glyph sits at the intersection of two different bubble boundaries, recognized as g by one projection system and as g' by another.

Toward Admissible Generative Typography

The framework developed in this part suggests a principled generative typography system distinct from existing approaches. Current generative and style-transfer systems optimize for statistical similarity to a training distribution, using objectives such as perceptual loss, feature matching, or discriminator feedback. These objectives do not explicitly model the admissibility bubble structure: a system optimizing perceptual loss can produce outputs near ∂B_g or even outside it without penalty, as long as the statistical features match.

An *admissible generative typography* system would instead explicitly maintain the constraint $\Phi(x) > \Phi_{\min}$ throughout generation, ensuring that every generated realization remains within the admissibility bubble. The RSVP field triple provides the architecture: the scalar field Φ serves as a learned recognizability density estimator; the vector field $\mathbf{v}$ defines the stylistic deformation flow for each typographic operation; the entropy field S controls the degree and direction of geometric disorder introduced at each step.

Define the generation process as a constrained variational problem:

$$\phi^* = \operatorname{argmax}_{\phi \in \mathcal{F}} \int_{X_{\text{stroke}}} [\mathcal{L}_{\text{style}}(x) - \mu \mathbf{1}[\Phi(\phi(x)) < \Phi_{\min}]] d\mu(x),$$

where $\mathcal{L}_{\text{style}}$ is the stylistic objective, $\mu > 0$ is a penalty weight for admissibility violations, and the constraint $\Phi(\phi(x)) \geq \Phi_{\min}$ enforces that generated realizations remain within the admissibility bubble.

The four typographic regimes — regular, bold, frenetic, hypnagogic — correspond to four different constraint geometries on this optimization. Regular maintains low S throughout; bold increases Φ by expanding stroke mass; frenetic maximizes $\text{Var}[\xi]$ subject to the entropy tangency constraint $(\nabla S)_\perp \approx 0$; hypnagogic maximizes the time spent near $\Phi_{\min}$ subject to the constraint that Φ oscillates above $\Phi_{\min}$ rather than falling below it.

The theoretical contribution of this framework is: it separates *semantic stability* from *geometric specificity*. A font can be arbitrarily geometrically novel — unprecedented stroke geometry, never-before-seen weight distribution, entirely original spatial frequency profile — while remaining semantically stable because it remains within the admissibility bubble. Conversely, a font that closely resembles a standard typeface in geometric parameters can be semantically unstable if its realizations concentrate near ∂B_g .

This separation is what allows function to survive collapse: the semantic invariant $\pi_{\text{glyph}}(x)$ is preserved not by preserving geometric fidelity but by remaining within the correct basin of attraction in the RSVP field geometry.

Conclusion: What Survives Collapse

Three results from distinct scientific domains have been examined in this essay. The first shows that anonymous additive aggregation suffices for universal secure computation, because local entropy injection can destroy provenance while preserving functional output. The second shows that bacterial gene expression is not a sequential pipeline but a geometry-constrained coordination of partially coupled flows, where mRNA fate depends on spatial accessibility rather than informational sequence. The third shows that early seed development in *Arabidopsis* is not a centrally orchestrated blueprint execution but a distributed constraint-propagation system in which sympastically isolated tissue compartments maintain developmental coherence through low-dimensional coordination operators across admissibility interfaces.

All three results displace the pipeline ontology from their respective domains. In computation, the pipeline from sender to receiver is replaced by local randomization plus additive aggregation. In prokaryotic gene regulation, the pipeline from DNA to protein is replaced by a partially coupled dynamical system in which spatial geometry, kinetic accessibility, and entropy management collectively determine which observables are stable. In plant development, the pipeline from genome to organism is replaced by a distributed field system in which tissue-

autonomous trajectories are coupled through sparse low-bandwidth signalling projections across constrained interfaces.

The admissible projection framework formalizes what all three results share: a map $\pi : X \rightarrow M$ that preserves computable functional observables while destroying microscopic recoverability. The mechanisms differ — mixing in the cryptographic case, topological separation in the bacterial case, sparse inter-manifold coordination in the developmental case — but the abstract structure is the same in each. None of the three systems preserves microscopic trajectory information. All three maintain functional invariants. They are realizations, through different physical substrates and at different scales, of the impossibility of local-to-global reconstruction in a constrained topological setting and of the persistence of function despite that impossibility.

The RSVP field triple $(\Phi, \mathbf{v}, S)$ provides the overarching dynamical geometry, with the admissibility conservation equation $\partial_t \Phi + \nabla \cdot (\Phi \mathbf{v}) = -\lambda S$ expressing the fundamental tension between trajectory flow and entropy production. Functional persistence requires that entropy flow remain tangent to the admissible manifold. When it does, entropy is a resource that enlarges fibers and deepens robustness. When it does not, entropy is a perturbation that drives trajectories out of the admissible regime and collapses function.

The deeper implication is a reorientation of scientific ontology. Classical systems theory asks how trajectories are preserved. The admissible projection framework asks which invariants survive trajectory destruction. That inversion is not a reformulation. It is a recognition that the systems most worth understanding — biological regulation, secure computation, developmental coordination, cognitive abstraction, morphogenesis — are precisely those in which trajectories are not preserved, and yet function persists anyway. Stability belongs not to the microscopic realization but to the projection class itself.

The microscopic paths are irretrievable. The functional output is stable. The projection is irreversible. The invariant persists.

The pipeline breaks. The invariant holds.

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